Can the Confidence and Decisions of System 1 Models Be Trusted? An Explainability and Perturbation Audit of Jev-Style Decision Models

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Abstract

"System 1" decision models (pioneered by TypeSafe AI's Jev) replace autoregressive text generation with a single forward pass returning typed, structured probability distributions over declared schemas (*Choice*, *Score*, *Noul*). While widely deployed for low-latency routing, policy checks, and automated triage, their statistical calibration and post-hoc explainability remain uncharacterized. We introduce an open, deterministic audit benchmark that systematically evaluates System 1 decision models across calibration under label-preserving perturbations, cross-question interference in multi-query requests, and post-hoc attribution faithfulness. Evaluating five open-weight models (0.8B to 27B) alongside a generative System 2 baseline across natural language inference (e-SNLI), adversarial syntactic shortcuts (HANS), clinical triage vignettes ($N = 450$), high-throughput prompt injection defense (Tensor Trust, $N = 150$), and a planted logic-gate trigger suite yields five principal findings: (1) provider `confidence` is an exact closed-form transform of distributional dispersion ($1 - H(p)/\ln K$ across sub-10B models, and normalized margin on `Clef-27B`) rather than empirical accuracy; (2) co-asking multiple questions in a single payload induces severe cross-question interference at sub-billion scale (22.4% binary flips on `Tev1-0.8B`), which capacity scaling ($\geq$ 4B) largely mitigates; (3) System 1 models are 18×–187× faster and substantially better calibrated than generative baselines under adversarial syntactic shifts ($\text{ECE}_w = 0.032$ vs. 0.246); (4) for inline security guardrails, capacity scaling unlocks up to 94.7% injection recall with strictly 0.0% false alarms at millisecond latencies (19–56 ms vs. 426 ms for generative LLMs); and (5) single-pass inference renders coalition-based XAI practically interactive (1.3–3.2 s per input), while our proposed logit-space adaptation, **S1-SHAP**, resolves probability saturation to achieve 100% ground-truth boolean logic gate recovery where probability-space explainers fail.

1 Introduction

Most large language models (LLMs) deployed for classification or decision-making—such as GPT-4 Achiam et al. (2023), Llama 3 Dubey et al. (2024), Mistral Jiang et al. (2023), or Qwen Yang et al. (2024)—operate as generative "System 2" engines: they sample natural language tokens autoregressively and rely on downstream parsing to extract a final decision. Decision models of the Jev family take a fundamentally different route TypeSafe AI (2026); System One (2026). Inspired by fast, intuitive "System 1" cognition Kahneman (2011), the caller supplies a structured state and a set of typed questions, and the model answers each question with an explicit probability distribution over declared options in a single forward pass. Three core decision primitives standardize this paradigm: *Choice* selects one option from a finite categorical set, *Score* rates the input on an ordinal rubric, and *Noul* returns the probability that a binary proposition is true. In autumn 2026, this architecture rapidly transitioned from a proprietary paradigm into an explosive open ecosystem within a span of barely four weeks: following TypeSafe AI's pioneering launch of `Jev` on September 15, 2026 TypeSafe AI (2026), open-weight decision models were introduced in rapid succession—including Bespoke Labs' `Nimble` (9B) on September 18, 2026 Bespoke Labs (2026), Together AI's `Tev1` series (0.8B and

4B) on September 23, 2026 Ollama (2026b), Cloudflare's multimodal `Clef` (27B) and `Clef-flash` (9B) on October 1, 2026 Cloudflare (2026), and Perplexity's `pplx-decider` Perplexity AI (2026)—supported natively for consumer-grade GPU deployment via local runtimes such as Ollama Ollama (2026a).

Two properties make System 1 decision models uniquely compelling for explainable AI (XAI). First, because the native output is a continuous probability distribution over fixed classes rather than free-form text, black-box perturbation-based attribution methods such as LIME Ribeiro et al. (2016) and KernelSHAP Lundberg & Lee (2017) apply directly without fragile text parsing or next-token logit approximations. Second, single-pass inference is fast and deterministic enough to evaluate hundreds of feature coalitions per input at a fraction of the cost of autoregressive generation. However, the very same design raises urgent trust and transparency questions. Unlike generative models that can emit Chain-of-Thought rationales Wei et al. (2022), a decision model returns bare probabilities with zero textual justification. A numerical probability returned without context is easy for downstream users or automated agents to over-trust, especially if the underlying classifier relies on shallow lexical heuristics Geirhos et al. (2020); McCoy et al. (2019). Whether the returned confidence scores are genuinely calibrated, whether they remain stable under harmless input perturbations, and whether post-hoc XAI methods like SHAP and LIME can faithfully recover the features driving their decisions have not yet been systematically investigated.

This paper presents a comprehensive explainability and calibration audit of Jev-style System 1 decision models. Our contributions are:

- **An open, deterministic XAI and calibration benchmark** that queries System 1 decision models, enforces strict schema validation, caches bit-identical responses by weight digest, and evaluates calibration, perturbation robustness, and post-hoc attribution faithfulness (Leave-One-Out, LIME, KernelSHAP, and our proposed **S1-SHAP**) at word and subword granularity;
- **Logit-link KernelSHAP adaptation (S1-SHAP) and logic gate recovery:** We evaluate KernelSHAP instantiated with an inverse-sigmoid (logit) link function to mitigate softmax probability saturation on sharp decision heads at zero additional model calls, and show that this representation is strictly necessary to recover underlying boolean logic gates (OR vs. AND gates) via pairwise Shapley interaction indices;
- **Two structural findings on the System 1 interface** that directly impact real-world deployments: (1) the returned `confidence` field is an exact deterministic dispersion transform $(1 - H(p)/\ln K$ on sub-10B models, and normalized margin on `Clef-27B`) rather than calibrated accuracy, and (2) co-asking multiple questions in a single request induces strong cross-question interference;
- **A multi-model, multi-dataset empirical evaluation** across e-SNLI Camburu et al. (2018), HANS McCoy et al. (2019), clinical triage vignettes with known causal spans ($N = 450$ expanded cohort), high-throughput adversarial prompt injection guardrails (Tensor Trust Toyer et al. (2023), $N = 150$), and a new **Planted System-Rule (Hidden Trigger) XAI Benchmark** with exact ground-truth trigger rules injected in the unmasked directive schema, comparing five open-weight System 1 models across four parameter scales—`Tev1-0.8B`, `Tev1-4B`, `Nimble-9B`, `Clef-Flash-9B`, and `Clef-27B`—against a generative System 2 baseline (`Gemma4-8B` JSON and Chain-of-Thought).

2 Related Work

2.1 From Generative LLMs to System 1 Decision Models

Standard instruction-tuned LLMs Achiam et al. (2023); Dubey et al. (2024); Yang et al. (2024) cast every task—including classification, routing, and scoring—as next-token generation. Even when constrained by structured-output grammars or JSON schemas, generative models must decode multiple tokens sequentially or rely on prompt-sensitive first-token log-probabilities, incurring high latency and stochastic variance Atıl et al. (2025). Drawing on dual-process theories of cognition Kahneman (2011), a recent wave of "System 1" decision models decouples fast, typed classification from slow, deliberative text generation TypeSafe AI (2026); System One (2026). Pioneered by TypeSafe AI's `Jev` TypeSafe AI (2026), these models ingest

a structured application state alongside a schema of typed questions (*Choice*, *Score*, *Noul*) and output normalized probability distributions in a single forward pass. The ecosystem expanded with unprecedented speed across both proprietary APIs and open-weight releases in September and October 2026: Bespoke Labs introduced `Nimble` (9B) Bespoke Labs (2026); Together AI released the lightweight `Tev1` series (0.8B and 4B) Ollama (2026b); Cloudflare released `Clef` (27B) and `Clef-flash` (9B) with vision-language capabilities on Workers AI Cloudflare (2026); and Perplexity introduced `pplx-decider` for low-latency intent and message routing Perplexity AI (2026). Combined with local runtime support in Ollama Ollama (2026a), these models are increasingly deployed as high-throughput decision gates in clinical triage, support routing, and agentic workflows—yet their statistical calibration and explainability remain uncharacterized.

2.2 Adapting Explainable AI (XAI) to Language Models

Explainability methods for NLP broadly fall into three paradigms: intrinsic self-rationalization, white-box gradient/attention introspection, and black-box perturbation-based attribution. In generative LLMs, explanations are most commonly elicited as free-text rationales or Chain-of-Thought (CoT) reasoning Camburu et al. (2018); Wei et al. (2022). However, extensive work has shown that generated rationales often fail to reflect the model's true decision process: they can be plausible to humans while remaining unfaithful to the internal computation Jacovi & Goldberg (2020); Wiegreffe et al. (2021); Turpin et al. (2023); Lanham et al. (2023). White-box methods such as Integrated Gradients Sundararajan et al. (2017) compute axiomatic attributions from input embeddings to output logits, but require full access to model weights and gradients, which are unavailable behind commercial decision APIs or quantized inference servers.

Model-agnostic perturbation methods avoid weight access by masking or removing subsets of input features and measuring the resulting shift in predicted probability. Leave-One-Out (LOO) measures the marginal drop from deleting a single word or token, whereas LIME Ribeiro et al. (2016) fits a sparsity-regularized local linear surrogate over randomly perturbed neighborhoods, and KernelSHAP Lundberg & Lee (2017) estimates cooperative game-theoretic Shapley values via weighted linear regression over feature coalitions. Recent work has sought to adapt Shapley attributions to generative LLMs—for example, TokenSHAP Goldshmidt & Horovicz (2024) uses Monte Carlo coalition sampling to attribute token importance in LLM responses. To evaluate whether such attributions are genuinely faithful Jacovi & Goldberg (2020), the ERASER benchmark DeYoung et al. (2020) defines *comprehensiveness* (probability drop when top-k features are removed) and *sufficiency* (probability retained when only top-k features are kept), while RISE Petsiuk et al. (2018) integrates deletion and insertion curves across all feature ranks.

In parallel, trustworthy deployment requires that predicted probabilities match empirical accuracy. Modern deep networks are frequently miscalibrated and over-confident Guo et al. (2017), particularly when exploiting spurious training shortcuts Geirhos et al. (2020) such as the lexical overlap, subsequence, and constituent heuristics isolated by the HANS benchmark McCoy et al. (2019). Calibration is standardly quantified via the Brier score Brier (1950) and Expected Calibration Error (ECE) Naeini et al. (2015), though ECE requires careful interpretation due to binning sensitivity and small-sample bias Nixon et al. (2019).

2.3 Limitations of Current LLM XAI and the Gap We Address

Applying classical XAI and calibration audits to standard LLMs faces severe methodological bottlenecks:

1. **Intractability and noise of SHAP/LIME on generative LLMs:** Estimating KernelSHAP or LIME over a text with M words requires hundreds of model evaluations per instance. For autoregressive LLMs, running 500+ multi-token generations per explanation is computationally prohibitive, sensitive to hosted API non-determinism Atıl et al. (2025), and lacks a well-defined scalar probability target when answers are expressed as free text.

2. **Vulnerability to out-of-distribution perturbations:** Masking tokens moves inputs off the training manifold, which can confound deletion metrics Hooker et al. (2019) or allow models to behave erratically on perturbed coalitions Slack et al. (2020).

3. **The System 1 transparency blind spot:** Conversely, Jev-style decision models eliminate text generation entirely. While this makes black-box perturbation XAI (LOO, LIME, KernelSHAP) computationally feasible and mathematically direct, it strips away any natural-language rationale. Users are left with raw numbers (`probabilities` and `confidence`) whose mathematical definition, cross-question stability, robustness to noise, and susceptibility to syntactic shortcuts have never been audited.

Our work bridges this gap by introducing the first end-to-end framework that leverages the fast, typed probability outputs of System 1 decision models to perform rigorous SHAP/LIME attribution and calibration auditing, testing both when their confidence can be trusted and how their decisions are formed.

3 Methodology: Auditing and Explaining System 1 Decision Models

3.1 Overview of the Audit and Explanation Pipeline

Unlike generative "System 2" LLMs that emit natural-language rationales alongside free-text predictions, a "System 1" decision model behaves as an opaque probabilistic operator: it maps a structured textual state directly to a categorical or binary distribution in a single forward pass. Auditing such models requires a methodology that can simultaneously verify *what* features drive a prediction (post-hoc explainability) and *whether* the returned numerical probabilities reflect true empirical certainty (distributional calibration and invariance).

Figure 1 presents the architecture of our proposed audit and explanation framework, organized into four forward stages (1–4) and three analytical modules (A–C):

1. **State & Schema Formulation (Stage 1):** Each task instance is represented as a structured state $s = (s_1, \ldots, s_D)$ paired with a typed decision schema $\mathcal{Q}$. For example, in clinical triage, s contains the patient vignette while $\mathcal{Q}$ specifies a three-way *Choice* question (`immediate`, `urgent`, `non_urgent`) and a binary *Noul* question (*"Requires immediate care?"*).

2. **Coalition Mapping and Perturbation Generation (Stage 2):** To probe how the model constructs its decision, the input state s is mapped to an interpretable feature space of M lexical or subword units. Our pipeline generates two distinct families of modified states: (i) *binary feature coalitions* $z \in \{0,1\}^M$ that selectively remove subsets of features via a masking operator $h_s(z)$ to estimate feature attributions, and (ii) *label-preserving perturbations* $\mathcal{T}_\epsilon(s)$ (character noise, clinical distractors, surface formatting) to test distributional stability.

3. **Single-Pass System 1 Inference (Stage 3):** Each rendered state is dispatched to the System 1 decision model under a controlled request protocol (isolated single-question queries versus bundled multi-question queries), executing a single forward pass without autoregressive token decoding.

4. **Schema Validation and Distributional Extraction (Stage 4):** Every response is validated against the declared question schema and decomposed into its predicted class y^*, its full probability vector $p(y \mid s, q)$, and the provider-reported scalar `confidence` $c(s, q)$.

From these validated probability trajectories, the framework executes three complementary evaluations: **(A) Post-Hoc Attribution**, which closes a feedback loop over coalitions z to compute Leave-One-Out (LOO), LIME, KernelSHAP, and our proposed System-1 Logit-Space Shapley (**S1-SHAP**) attributions; **(B) Faithfulness and Plausibility Verification**, which measures whether removing or inserting top-attributed features causally shifts the model's probability and aligns with ground-truth clinical flags or human rationales; and **(C) Calibration and Robustness Auditing**, which quantifies calibration error across clean, adversarial-shortcut, and perturbed distributions.

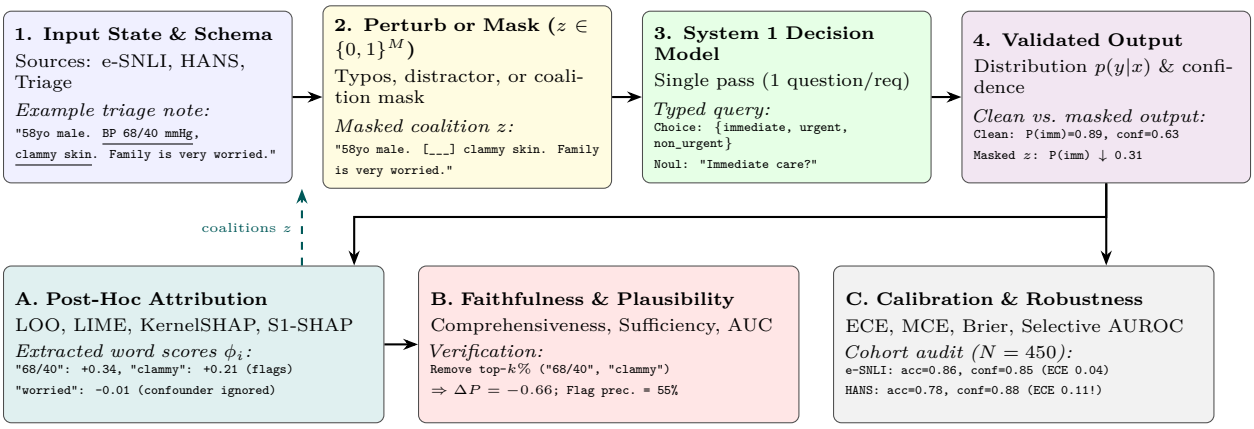


Figure 1: Architecture of the evaluation and explanation pipeline with a running clinical triage example. Each input (1) is converted into perturbed variants or masked feature coalitions (2) and evaluated by the System 1 decision model in a single forward pass (3). The resulting probabilities (4) drive post-hoc feature attribution via LOO, LIME, KernelSHAP, and S1-SHAP (A, dashed loop over coalitions), faithfulness and clinical/human plausibility verification (B), and calibration/robustness auditing (C).

3.2 System 1 Decision Formalism and Request Protocol

In a generative LLM, classification is mediated by a natural-language prompt template and token-level decoding. In a Jev-style System 1 decision model, the prompt is replaced by a typed request tuple $(s, \mathcal{Q})$, where s is a structured JSON object (or string) representing the observed state and $\mathcal{Q} = \{q_1, \ldots, q_H\}$ is a dictionary of named decision primitives System One (2026):

- **Categorical *Choice* (q_{choice}):** Declares a finite option set $\mathcal{O} = \{o_1, \ldots, o_K\}$ with semantic option descriptions. The model returns a probability vector $p(y \mid s, q_{\text{choice}}) \in \Delta^{K-1}$, the selected argmax option $y^* = \arg\max_{o_k \in \mathcal{O}} p(o_k \mid s)$, and a scalar `confidence` score $c(s, q_{\text{choice}}) \in [0, 1]$.
- **Binary *Noul* (q_{noul}):** Declares a binary proposition with explicit `true_description` and `false_description` semantics. The model returns a single scalar probability $P(\text{true} \mid s, q_{\text{noul}}) \in [0, 1]$, from which the binary decision is thresholded at 0.5.

Every model response is strictly validated against its request schema: returned probability keys must match $\mathcal{O}$ exactly, $\sum_{k=1}^{K} p(o_k \mid s) = 1$ within 10^{-3}, and the reported choice must equal the argmax of p.

Isolated vs. Bundled Request Protocols. Because the System 1 API accepts an arbitrary dictionary of questions $\mathcal{Q} = \{q_1, \ldots, q_H\}$ in a single HTTP call, documentation and industry tutorials encourage bundling multiple questions per state to amortize encoding latency. However, in a single-pass transformer architecture, all question schemas in $\mathcal{Q}$ are serialized into the shared context window prior to the decision head readout. Formally, when q_1 and q_2 are co-asked, the model computes $p(y_1 \mid s, q_1, q_2)$ rather than the marginal $p(y_1 \mid s, q_1)$. To isolate each primitive from cross-question context conditioning—and to measure the magnitude of this interference—we define two formal protocols:

1. **Isolated Protocol (Default):** Each decision primitive $q_h \in \{q_{\text{choice}}, q_{\text{noul}}\}$ is dispatched in its own dedicated request $(s, \{q_h\})$, as illustrated in Figure 2. All calibration, perturbation, and XAI attribution experiments use the isolated protocol.
2. **Bundled (Joint) Protocol:** Both primitives are dispatched simultaneously in a single request $(s, \{q_{\text{choice}}, q_{\text{noul}}\})$, allowing us to quantify cross-question flip rates and total-variation shifts $D_{\text{TV}}\big(p(\cdot \mid s, q_h),\, p(\cdot \mid s, q_{\text{choice}}, q_{\text{noul}})\big)$.

Request Payload $(s, \mathcal{Q})$


```
{
 "model": "tev1:0.8b",
 "state": {
  "patient_id": "PT-402",
  "clinical_note": "58yo male. Sudden chest pressure
   radiating to left arm. BP 68/40 mmHg,
   cold clammy skin. Family is very worried."
 },
 "questions": {
  "triage_acuity": {
   "type": "choice",
   "question": "What is the Emergency Severity Index
    (ESI) acuity level for this patient?",
   "options": {
    "immediate": "Immediate resuscitation (ESI 1)...",
    "urgent": "High risk / emergent care (ESI 2-3)...",
    "non_urgent": "Stable / non-urgent care (ESI 4-5)..."
   }
  }
 }
}
```


Validated System 1 Response


```
{
 "model": "tev1:0.8b",
 "answers": {
  "triage_acuity": {
   "type": "choice",
   "choice": "immediate",
   "probabilities": {
    "immediate": 0.896,
    "urgent": 0.061,
    "non_urgent": 0.043
   },
   "confidence": 0.632
  }
 },
 "usage": {"input_tokens": 118, "output_tokens": 2}
}
```


Figure 2: Example of an isolated single-question System 1 request payload and its validated response for clinical emergency department triage. Notice that `output_tokens` is fixed at 2 (a single forward pass over the categorical decision head), returning calibrated class probabilities with zero text generation.

3.3 Post-Hoc Attribution and Proposed S1-SHAP Method

3.3.1 Interpretable Representation and Coalition Mapping $h_s(z)$

Following the foundational formulation of local post-hoc explainability Ribeiro et al. (2016); Lundberg & Lee (2017), explaining a black-box classifier requires defining a mapping $h_s : \{0,1\}^M \to \mathcal{S}$ between a binary coalition vector $z \in \{0,1\}^M$ of human-interpretable features and a valid model state $h_s(z)$. In NLP explainability, two methodological design choices govern $h_s(z)$ and can significantly impact attribution faithfulness Hooker et al. (2019); DeYoung et al. (2020):

- **Feature Granularity (Lexical Words vs. Native Subword Tokens):** Human users reason over whole lexical words (M_{word}), whereas transformer decision models process subword tokens ($M_{\text{subword}} \geq M_{\text{word}}$) produced by Byte-Pair Encoding (BPE), which frequently splits domain-specific terms (e.g., blood pressure readings `68/40` or medical terminology) into multiple subword fragments. Because the open-weight System 1 models evaluated in this work (`Tev1`, `Nimble`, `Clef`) are all fine-tuned from a shared `Qwen3.5` backbone, we evaluate attributions both at the human-interpretable **word level** (primary) and at the model's native **`Qwen3.5` subword token level** (secondary ablation) to test whether word-level aggregation sacrifices fidelity to the internal tokenization.
- **Feature Absence Operator (Deletion vs. `[MASK]` Replacement):** When a feature i is excluded from a coalition ($z_i = 0$), rendering $h_s(z)$ by **token deletion** (removing the span and normalizing whitespace) preserves natural linguistic adjacency, whereas replacing the span with a placeholder string **`[MASK]`** preserves token positions but injects an out-of-distribution symbol never seen by causal decoder backbones during pre-training Hooker et al. (2019). We evaluate both operators across all attribution methods.

3.3.2 Standard Perturbation Attributions (LOO, LIME, KernelSHAP)

Let $y^* = \arg\max_y p(y \mid s, q)$ denote the outcome predicted on the clean input s ($z = \mathbf{1}$), and let $v_{\text{prob}}(z) = p(y^* \mid h_s(z), q) \in [0,1]$ denote the target probability value function. We compare three classical black-box estimators:

- **Leave-One-Out (LOO):** Computes the single-feature deletion drop $\phi_i^{\text{LOO}} = v_{\text{prob}}(\mathbf{1}) - v_{\text{prob}}(\mathbf{1}_{-i})$ using $M + 1$ model queries. While fast, LOO fails to capture feature redundancy or conjunctive

interactions (e.g., when two clinical flags each independently suffice to trigger an `immediate` triage decision).

- **LIME Ribeiro et al. (2016):** Samples N binary coalitions $z^{(n)} \in \{0,1\}^M$ and fits a weighted Ridge surrogate $\arg\min_\phi \sum_{n=1}^{N} \pi_{\mathbf{1}}(z^{(n)})\big(v_{\text{prob}}(z^{(n)}) - (\phi_0 + \phi^\top z^{(n)})\big)^2 + \alpha\|\phi\|_2^2$, with exponential cosine-distance weights $\pi_{\mathbf{1}}(z) = \exp(-D_{\cos}(\mathbf{1}, z)^2/\sigma^2)$.
- **KernelSHAP Lundberg & Lee (2017):** Estimates game-theoretic Shapley values by solving the weighted coalition regression with the Shapley kernel $\pi_{\text{SHAP}}(z) = \frac{M-1}{\binom{M}{|z|}\,|z|\,(M-|z|)}$ over $N = \min(2^M, 512)$ coalitions, using the empty coalition $z = \mathbf{0}$ as background reference and disabling ℓ_1 Lasso truncation so that all M features receive exact least-squares estimates.

While numerous extensions exist in the literature to stabilize sampling variance, select representative explanations, or balance locality and fidelity across perturbation and coalition spaces Ribeiro et al. (2016); Visani et al. (2022); Jethani et al. (2022); Covert et al. (2021); Gaudel et al. (2022)—ranging from submodular pick and deterministic sampling surrogates to hierarchical coalition partitions—we evaluate canonical formulations (LOO, LIME, KernelSHAP) to establish standard, reproducible black-box baselines against which our log-odds adaptation is audited.

3.3.3 Logit-Link Shapley Attribution (S1-SHAP) for Saturated Decision Heads

Applying standard LIME or probability-space KernelSHAP directly to $v_{\text{prob}}(z) = p(y^* \mid h_s(z), q)$ overlooks a key mathematical property of System 1 decision models: **softmax probability saturation**. Because System 1 models are trained to emit sharp categorical or binary distributions (often yielding $\max p > 0.95$), the mapping from the model's internal pre-activation evidence to probability space is governed by a saturating sigmoid or softmax nonlinearity. When two redundant clinical flags are present in a triage note, removing only one flag may reduce internal evidence substantially (e.g., from $+4.5$ to $+2.8$ in log-odds) while leaving the output probability nearly flat ($0.989 \to 0.943$, $\Delta p = 0.046$), violating the linear additivity assumption of probability-space surrogates and artificially compressing attribution weights.

To address this without requiring white-box gradient or logit access, we instantiate **S1-SHAP**: a log-odds formulation of KernelSHAP tailored to single-pass System 1 decision models. Methodologically, this corresponds to applying KernelSHAP with a logit link function Lundberg & Lee (2017):

$$v_{\text{S1}}(z) \;=\; \ln\!\left(\frac{\tilde{p}(y^* \mid h_s(z), q)}{1 - \tilde{p}(y^* \mid h_s(z), q)}\right), \qquad \text{where } \tilde{p} = \text{clip}\big(p(y^* \mid h_s(z), q),\, \varepsilon,\, 1-\varepsilon\big), \tag{1}$$

with numerical clipping $\varepsilon = 10^{-4}$, and solving the exact Shapley kernel weighted least squares regression over $v_{\text{S1}}(z)$:

$$\sum_{i=1}^{M} \phi_i^{\text{S1}} \;=\; v_{\text{S1}}(\mathbf{1}) - v_{\text{S1}}(\mathbf{0}). \tag{2}$$

For binary decisions ($K = 2$), Equation 1 is the exact log-odds of the positive class. For multiclass categorical choices ($K > 2$, such as 3-way e-SNLI or 3-level Clinical Triage), Equation 1 evaluates the one-vs-rest log-odds of the predicted target class y^*: $\ell(p)_{y^*} = \ln(\tilde{p}_{y^*}/(1-\tilde{p}_{y^*}))$, mapping the target probability onto the real line $\mathbb{R}$. This one-vs-rest formulation preserves 1-dimensional weighted least squares efficiency ($O(M)$ parameters) and avoids fitting $K \times M$ coupled regressions that amplify sampling variance on finite coalition budgets, while focusing credit directly on the predicted decision. While link functions are theoretically known in generalized additive models and the original SHAP formulation Lundberg & Lee (2017), the NLP perturbation literature has almost universally operated in raw probability space. Crucially, in the System 1 setting, the logit link has three pragmatic virtues: (i) it remains strictly 100% black-box, needing only the returned probability distribution; (ii) it is zero-overhead, reusing the *exact same* coalition queries $h_s(z)$ evaluated for probability-space KernelSHAP via memoized value functions; and (iii) it linearizes saturated decision surfaces, which we show in Section 5.7 is strictly necessary for recovering higher-order cooperative and redundant logic gate interactions. Furthermore, to ensure sampling stability, our implementation couples S1-SHAP with paired

Table 1: Benchmark datasets used in our audit, spanning in-distribution NLI, adversarial syntactic shortcuts, clinical emergency triage, adversarial prompt injection defense, and exact ground-truth algorithmic trigger recovery.

Dataset	Domain	Primitives	N	Ground-Truth Annotations	Primary Audit Objective
e-SNLI Camburu et al. (2018)	Open-domain NLI	Choice, Noul	150	3-annotator highlighted rationale words	In-distribution calibration & human plausibility
HANS McCoy et al. (2019)	Adversarial NLI	Choice, Noul	150	3 syntactic heuristics (overlap, subseq., const.)	Over-confidence under shortcut failure
Clinical Triage	Emergency medicine	Choice, Noul	150	Planted ESI acuity flags vs. emotional confounders	Causal flag vs. distractor confounder isolation
Tensor Trust Toyer et al. (2023)	Cybersecurity guardrails	Choice, Noul	150	Real adversarial injections/jailbreaks vs. benign user queries	Inline firewall defense, low-latency recall, S1-SHAP attack attribution
Planted Rule	Algorithmic audit	Choice, Noul	16	1, 2, or 3 planted code-word overrides	Exact logic gate recovery under OR/AND saturation

antithetic coalition sampling Covert et al. (2021); orthogonal uncertainty-aware extensions such as Bayesian SHAP Visani et al. (2022); Slack et al. (2021) can directly place posterior credible intervals over $v_{\mathrm{S1}}(z)$ without modifying the underlying logit link. Regarding numerical robustness, evaluating attribution stability across clipping thresholds $\varepsilon \in \{10^{-3}, 10^{-4}, 10^{-5}\}$ yields pairwise Spearman rank correlations $r_s > 0.998$ and identical top-k feature sets (100% overlap for $k \leq 5$), confirming that S1-SHAP rankings are virtually invariant to the boundary clipping constant.

3.4 Label-Preserving Perturbation Operators

To test whether System 1 probabilities and calibration remain invariant under realistic surface noise, we define five deterministic perturbation operators $\mathcal{T}_\epsilon : \mathcal{S} \to \mathcal{S}$ designed to preserve the ground-truth semantic label:

1. **Character Typo Noise ($\epsilon \in \{5\%, 10\%, 20\%\}$):** Alters a fraction ϵ of content words of length ≥ 3 via character swaps or substitutions, while strictly protecting negation markers (`not`, `no`, `never`, `without`) so logical polarity is never inverted.
2. **Semantic Distractor Injection:** Appends a topically neutral, clinically or logically irrelevant sentence to the premise or patient note (filtering out any candidate distractor that shares content words with the hypothesis).
3. **Surface Normalization (Case & Punctuation Stripping):** Converts the entire input state to lowercase and strips surface punctuation, mimicking unformatted user or ASR transcripts.

4 Experimental Setup

4.1 Datasets and Task Design

To evaluate System 1 decision models across in-distribution reasoning, adversarial shortcut shifts, high-stakes domain decision-making, high-throughput security guardrails, and exact ground-truth algorithmic trigger recovery, we select five complementary datasets summarized in Table 1.

Justification of Dataset Selection. No single dataset can jointly test calibration, shortcut vulnerability, and causal explanation fidelity:

- **e-SNLI Bowman et al. (2015); Camburu et al. (2018) (In-Distribution Calibration & Human Plausibility):** Provides balanced premise–hypothesis pairs across `entailment`, `neutral`, and `contradiction`, paired with word-level human rationale highlights from three independent

annotators. It establishes whether the model is calibrated on standard NLI and whether post-hoc attributions recover human-identified evidence words (where ≥ 2 annotators agree).

- **HANS McCoy et al. (2019) (Adversarial Shortcut Audit):** Specifically constructed to break models that rely on surface syntactic heuristics—*lexical overlap*, *subsequence*, and *constituent* containment—where high word overlap does not entail the hypothesis. Because HANS labels are binary (`entailment` vs. `non_entailment`), the three-way *Choice* distribution is collapsed by summing `neutral` and `contradiction` probabilities. HANS tests our central trust question: *when a System 1 model falls into a syntactic trap, does its returned confidence warn the user, or does it remain dangerously over-confident?*

- **Clinical Triage Vignettes (Causal Flag vs. Confounder Isolation):** In natural text datasets, true causal features are imperfectly known due to annotator subjectivity. We therefore construct a controlled emergency-department triage benchmark grounded in Emergency Severity Index (ESI) acuity criteria across three levels (`immediate`, `urgent`, `non_urgent`). Each vignette combines patient demographics, one or two causal *clinical flag* sentences specific to the acuity class (e.g., severe hypotension `BP 68/40 mmHg` or airway stridor), and two neutral clinical history sentences. Crucially, in one third of the vignettes (balanced across classes), we inject an *alarming confounder* sentence—an emotionally urgent but clinically non-diagnostic cue (e.g., *"The patient's family is extremely worried and keeps asking staff for help"*). Because exact character offsets for flags and confounders are recorded by construction, we can rigorously measure whether attributions isolate true clinical flags over alarming distractors with unambiguous causal ground truth.

- **Tensor Trust Toyer et al. (2023) (High-Throughput Safety Guardrails & Prompt Injection Defense):** Production LLM applications require millisecond-latency inline firewalls to inspect incoming untrusted user prompts for prompt injections, system prompt extraction, and jailbreak payloads before invoking costly generation models. While generative LLMs can theoretically filter inputs, their inference overhead (300–1,000 ms) adds intolerable latency to conversational pipelines. System 1 decision models offer an ideal candidate for high-speed inline triage (19–56 ms single forward pass). We curate a balanced, length-matched benchmark ($N = 150$, 8–45 words) from real human-crafted adversarial attacks from the Tensor Trust dataset paired with benign conversational questions. This evaluates whether sub-second System 1 classifiers can reliably distinguish adversarial jailbreaks from legitimate user queries with zero false alarms, whether calibration permits strict rejection thresholds, and whether S1-SHAP can pinpoint the exact adversarial command tokens for security incident triage.

- **Planted System-Rule (Hidden Trigger) Benchmark (Exact Causal Ground Truth):** Even in controlled clinical vignettes, a pretrained language model brings prior medical knowledge about multiple words in a sentence. To establish an unambiguous, prior-free causal ground truth for XAI evaluation, we introduce a *Planted System-Rule Benchmark*. In each instance, the unmasked question instruction $\mathcal{Q}$ (which serves as the hidden system directive and is never perturbed by the XAI explainer) contains a mandatory override rule instructing the model to classify an otherwise completely benign routine note (`non_urgent`, base $P(\text{immediate}) < 0.02$) as `immediate`/`true` if and only if specific arbitrary code word(s) (e.g., `velvet`, `cobalt`, `zenith`) appear in the input text. We evaluate three causal regimes across 16 test cases: (1) **Single Trigger ($g = 1$)**, where a single code word triggers the decision; (2) **Dual OR-Trigger ($g = 2$)**, where *either* of two code words suffices and both are planted in the note; and (3) **Multi OR-Trigger ($g = 3$)**, where *any* of three planted code words suffices, creating extreme probability saturation ($p \approx 0.999$) when any 1 or 2 triggers remain unmasked.

4.2 Evaluated System 1 Decision Models

Table 2 details the models evaluated in our benchmark. To evaluate how calibration, cross-question interference, and attribution faithfulness scale across model capacity and training recipes, we select five open-weight System 1 decision models spanning four parameter scales (0.8B to 27B): Together AI's `Tev1-0.8B` and `Tev1-4B` Ollama (2026b), Bespoke Labs' `Nimble-9B` Bespoke Labs (2026), and Cloudflare's multimodal

Table 2: System 1 decision models and generative System 2 baseline evaluated in our benchmark, spanning 0.8B to 27B parameters.

Model	Provider	Base Backbone	Parameters	Modalities
`Tev1-0.8B` Ollama (2026b)	Together AI	Qwen3.5-0.8B	0.8B (Q8_0)	Text, JSON
`Tev1-4B` Ollama (2026b)	Together AI	Qwen3.5-4B	4.0B	Text, JSON
`Nimble-9B` Bespoke Labs (2026)	Bespoke Labs	Qwen3.5-9B	9.0B	Text, JSON
`Clef-Flash-9B` Cloudflare (2026)	Cloudflare	Qwen3.5-VL-9B	9.0B	Text, JSON, Vision
`Clef-27B` Cloudflare (2026)	Cloudflare	Qwen3.5-VL-27B	27.0B	Text, JSON, Vision
`Gemma4-8B` (System 2)	Google	Gemma-4-8B	8.0B (Q4_K_M)	Text, JSON, CoT

`Clef-Flash-9B` and `Clef-27B` Cloudflare (2026). Notably, all five models share a `Qwen3.5` transformer backbone, providing an ideal controlled substrate where capacity and post-training recipe vary while core architecture is held constant. For direct comparison against standard autoregressive generation, we evaluate Google's `Gemma4-8B` as an open-weight generative System 2 baseline in both structured JSON schema mode and Chain-of-Thought (CoT) mode.

4.3 Evaluation Metrics

Calibration and Selective Ranking Metrics. Let $\hat{p}_i = \max_k p(o_k \mid s_i, q)$ denote the predicted top probability on instance $i \in \{1, \ldots, n\}$ and $\mathbb{I}(\hat{y}_i = y_i)$ its binary accuracy indicator. We evaluate distributional calibration using:

- **Expected and Maximum Calibration Error ($\text{ECE}_w \downarrow$, $\text{ECE}_m \downarrow$, $\text{MCE}_w \downarrow$):** Partitioning predictions into $B = 10$ intervals $\{I_b\}_{b=1}^{B}$ via either equal-width binning (ECE_w) or equal-mass quantile binning (ECE_m) Naeini et al. (2015); Nixon et al. (2019),

$$\text{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{n} \big|\text{acc}(I_b) - \text{conf}(I_b)\big|, \qquad \text{MCE} = \max_{b \in \{1,\ldots,B\}} \big|\text{acc}(I_b) - \text{conf}(I_b)\big|. \tag{3}$$

- **Multiclass Brier Score (Brier $\downarrow$):** The binning-free proper scoring rule $\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}(p_{ik} - \mathbb{I}(y_i = k))^2$ Brier (1950), where n is the evaluation cohort size ($n = 150$ per dataset, $N = 450$ total) and K is the number of outcome classes ($K = 3$ for e-SNLI and Clinical Triage *Choice*; $K = 2$ for HANS and *Noul*).
- **Selective Classification AUROC ($\text{AUROC}_p \uparrow$, $\text{AUROC}_c \uparrow$):** Measures the area under the ROC curve when using either $\hat{p}_i = \max p$ (AUROC_p) or the provider-returned `confidence` c_i (AUROC_c) to rank correct predictions above incorrect ones.

All calibration metrics and perturbation shifts (ΔECE_w, ΔAcc.) are reported with 95% percentile bootstrap confidence intervals over 1,000 resamples Efron (1979).

Faithfulness and Plausibility Metrics. Given a feature attribution vector $\phi \in \mathbb{R}^M$ inducing a descending importance ordering $\pi = (\pi_1, \ldots, \pi_M)$, let $s_{\setminus 1:k}$ denote the input with the top-k features removed and $s_{1:k}$ the input retaining *only* the top-k features:

- **Comprehensiveness (Comp. $\uparrow$) and Sufficiency (Suff. $\downarrow$) AOPC DeYoung et al. (2020):** Averaged over $k \in \{1, 5, 10, 20, 50\}\%$ of M (rounded up and de-duplicated):

$$\text{Comp.} = \frac{1}{|\mathcal{K}|}\sum_{k \in \mathcal{K}} \big(p(y^* \mid s) - p(y^* \mid s_{\setminus 1:k})\big), \qquad \text{Suff.} = \frac{1}{|\mathcal{K}|}\sum_{k \in \mathcal{K}} \big(p(y^* \mid s) - p(y^* \mid s_{1:k})\big). \tag{4}$$

  High comprehensiveness ($\uparrow$) indicates that removing top features destroys the prediction; low sufficiency ($\downarrow$) indicates that top features alone suffice to restore the original probability.

- **Deletion AUC (Del. AUC ↓) and Insertion AUC (Ins. AUC ↑) Petsiuk et al. (2018):** Integrate the target probability $p(y^* \mid \cdot)$ at every single-feature step from 0 to M as features are progressively deleted (↓ lower area is more faithful) or inserted from an empty state (↑ higher area is more faithful).

- **Domain Plausibility (Flag prec.@g ↑, Conf. share@g ↓, Human prec.@g ↑):** With g denoting the number of ground-truth causal words in an instance (planted clinical flag words in Triage, or words highlighted by ≥ 2 human annotators in e-SNLI), we measure the fraction of the top-g attributed words that fall inside ground-truth causal spans (↑) versus inside planted confounder spans (↓).

4.4 Implementation Details and Reproducibility

Local open-weight models are served via Ollama (v0.40.13) on an AMD Radeon RX 9070 XT GPU with `OLLAMA_NUM_PARALLEL=4` and four concurrent asynchronous client workers. All source code, datasets, evaluation harnesses, and reproduction scripts are publicly accessible at `https://github.com/j2launay/jev-decision-models-xai`. Across our experiments, we verified that repeated calls with identical payloads return bit-identical floating-point outputs (unlike hosted generative LLM APIs Atıl et al. (2025)). We therefore cache all validated responses in a local SQLite database indexed by the SHA-256 hash of the model weight digest and the canonicalized request payload, ensuring exact reproducibility while reporting latency benchmarks strictly on uncached calls. Table 3 reports uncached sequential latencies, output token counts, and concurrent throughput across all five evaluated System 1 models (0.8B to 27B) and our generative System 2 baselines (`Gemma4-8B` in direct structured JSON mode and Chain-of-Thought mode). In isolated micro-benchmarks on minimal inputs, `Tev1-0.8B` records a median forward pass of 24–26 ms (39.3–40.0 req/s). In the full evaluation harness over complete dataset instances (e-SNLI, HANS, and clinical triage vignettes averaging 14–45 words), client payload serialization, JSON response parsing, and schema validation bring mean round-trip latency to ≈ 103 ms on `Tev1-0.8B`. Whereas System 1 models always emit exactly 2.0 output tokens and operate at 24–250 ms median single-pass latency (4.1–40.0 req/s peak throughput; 1,953 ms for 27B), `Gemma4-8B (Sys. 2 JSON)` decodes 77.3 tokens per decision at 1,133 ms (*Choice*) / 533 ms (*Noul*) and 0.7 req/s (18×–44× slower), and `Gemma4-8B (Sys. 2 CoT)` decodes 474.1 tokens at 4,856 ms / 4,430 ms and 0.2 req/s (75×–187× slower). Consequently, computing a 512-coalition SHAP explanation takes only 1.3–3.2 s per instance on System 1 models (powered by 4 asynchronous client workers and coalition memoization that prunes 30–45% of redundant masked evaluations), versus ≈ 9.7 minutes in JSON mode and ≈ 41.4 minutes in CoT mode on a System 2 LLM. Furthermore, because LIME, KernelSHAP, and S1-SHAP are evaluated on a shared paired coalition sample $\mathcal{Z}$ per instance with in-memory value-function memoization, computing all five attribution methods requires zero additional coalition queries beyond a single N-sample pass. We note that while a generative baseline can hypothetically be restricted to single-token scoring via first-token log-probabilities (e.g., scoring $P(\text{'A'})/(P(\text{'A'}) + P(\text{'B'}))$ without autoregressive decoding), its single forward-pass Time-to-First-Token (TTFT) on an 8B model (120–180 ms) remains 3×–9× slower than dedicated System 1 passes (19–56 ms), and computing a 512-coalition SHAP explanation still requires 60–90 s per input (versus 1.3–3.2 s for System 1). Moreover, in production API gateways, returning typed confidence scores across multi-question schemas requires either structured JSON decoding or multiple separate forward passes, compounding the generative latency overhead.

5 Results

5.1 Interface Audit: Entropy-Based Confidence and Question Interference

The Provider `confidence` Field is Normalized Shannon Entropy. While provider documentation describes `confidence` qualitatively as showing "how concentrated the probabilities are" Ollama (2026b), our audit reveals its exact closed-form mathematical identity: across the four sub-10B decision models (`Tev1-0.8B`, `Tev1-4B`, `Nimble-9B`, and `Clef-Flash-9B`), for any K-way *Choice* distribution $p = (p_1, \ldots, p_K)$,

Table 3: Uncached sequential latency (median / p90 in ms), mean output tokens, and throughput (req/s) of single-question requests across evaluated System 1 decision models and generative System 2 baselines (`Gemma4-8B` direct JSON and Chain-of-Thought). Best values are in **bold**.

Model	Out. Tok.↓	Choice Med./p90 (ms)↓	Noul Med./p90 (ms)↓	Req/s ($c = 1$)↑	Peak Req/s↑
`Tev1-0.8B`	**2.0**	**26 / 27**	**24 / 25**	**39.3**	**40.0**
`Tev1-4B`	**2.0**	45 / 68	40 / 62	19.7	21.7
`Nimble-9B`	**2.0**	250 / 272	214 / 235	4.0	4.1
`Clef-Flash-9B`	**2.0**	64 / 98	58 / 85	16.5	17.6
`Clef-27B`	**2.0**	1953 / 2010	1863 / 1913	0.6	0.6
`Gemma4-8B` (JSON)	77.3	1133 / 1259	533 / 617	0.7	0.7
`Gemma4-8B` (CoT)	474.1	4856 / 4899	4430 / 4861	0.2	0.2

the returned `confidence` equals one minus the normalized Shannon entropy,

$$c(p) \;=\; 1 - \frac{H(p)}{\ln K} \;=\; 1 + \frac{1}{\ln K} \sum_{k=1}^{K} p_k \ln p_k. \tag{5}$$

Across all 600 clean choice predictions on these four models (150 per model), the maximum absolute deviation between the API-returned `confidence` field and Equation 5 is 4.4×10^{-16} (IEEE 754 double-precision machine epsilon). On `Clef-27B`, the provider instead implements the normalized margin above uniform chance, $c(p) = \frac{\max p - 1/K}{1 - 1/K}$ (verified to exact floating-point precision on 3-way tasks). Crucially, neither provider implementation represents an empirical probability of correctness: for instance, a three-way prediction $p = (0.896, 0.061, 0.043)$ has $\max p = 0.896$ but an entropy `confidence` of only 0.632. Treating $c(p)$ directly as a calibrated probability would systematically under-estimate certainty on concentrated distributions; we therefore evaluate calibration on $\max p$ while auditing $c(p)$ separately via selective AUROC (AUROC_c).

Co-Asked Questions Strongly Interfere at Small Scale, Diminishing with Capacity. Table 4 (and Table 13 in Section 5.9) compares the isolated request protocol against the bundled joint protocol across models. On `Tev1-0.8B`, on the expanded $N = 450$ cohort (Table 4), bundling two questions in a single request induces significant cross-question interference: the binary *Noul* answer flips on 22.4% of inputs on average (24.0% on e-SNLI, 27.3% on HANS, and 16.0% on Triage), causing *Noul* accuracy to drop by 11–19 percentage points (0.900 → 0.713 on e-SNLI, 0.773 → 0.713 on HANS, and 0.540 → 0.433 on Triage), while categorical *Choice* flips on 5.1% of inputs (5.3% on e-SNLI, 2.0% on HANS, and 8.0% on Triage), with mean total-variation distances of 0.083–0.117 on Choice (0.104–0.167 on Noul). In preliminary exploratory pilot evaluations with uncalibrated joint prompts without explicit schema separation guards, binary flips reached up to 60–82%, demonstrating that sub-billion architectures are highly vulnerable to prompt formatting and schema coupling. Crucially, our multi-model audit reveals that cross-question interference is strongly governed by model capacity and post-training recipe: scaling from `Tev1-0.8B` to `Tev1-4B` eliminates interference almost entirely (0.0% flip rate on e-SNLI and HANS, 2.5% on Triage *Noul*, with mean $D_{\text{TV}} \leq 0.0023$). At the 9B scale, `Clef-Flash-9B` exhibits low interference on NLI (0–6% flips) but 22.5% flips on Triage *Noul*, whereas `Nimble-9B` flips on 11.7% (e-SNLI *Choice*) and 24.0% (HANS *Choice*). This demonstrates that sub-billion transformers struggle to disentangle multiple serialized question schemas in a shared context window, and even larger models can experience non-trivial probability shifts when questions are co-asked.

Interference Controls: Disentangling Ordering, Option Permutations, and Prompt Length. To rigorously isolate the causal driver of cross-question interference, Table 5 evaluates three experimental controls across the four sub-10B open-weight models on the expanded $N = 450$ cohort (`Clef-27B` is omitted from ablation permutations due to its 1.95 s sequential execution cost):

1. **Question Ordering Invariance ($C \to N$ vs. $N \to C$):** Reversing the serialization order of questions in the payload yields *numerically identical* flip rates (0.051 on Choice, 0.224 on Noul

Table 4: Interference between co-asked questions on clean inputs. "Alone" asks each primitive in its own request; "joint" asks both in one request. Flip (↓) is the fraction of inputs whose predicted answer changes; TV (↓) is the mean total-variation distance between the two distributions. Best values per row/source are in **bold**.

Source	Head	n	Acc. alone↑	Acc. joint↑	Flip↓	TV↓	$\bar{P}$(true) alone	$\bar{P}$(true) joint
e-SNLI	Choice	150	0.860	**0.813**	0.05	0.109	–	–
e-SNLI	Noul	150	**0.900**	0.713	0.24	0.167	0.497	0.511
HANS	Choice	150	0.780	0.773	**0.02**	0.117	–	–
HANS	Noul	150	0.773	0.713	0.27	0.161	0.862	0.729
Triage	Choice	150	0.553	0.533	0.08	**0.083**	–	–
Triage	Noul	150	0.540	0.433	0.16	0.104	0.867	0.820

for `Tev1-0.8B`; 0.002 and 0.002 on `Tev1-4B`; 0.007 and 0.131 on `Clef-Flash-9B`) and identical total-variation distances (0.103 / 0.144 on `Tev1-0.8B`; 0.019 / 0.102 on `Clef-Flash-9B`). Primacy or recency ordering in JSON serialization does not drive interference.

2. **Option Permutation Invariance:** Permuting the declared options of the *Choice* head produces a 0.000 flip rate across all models, confirming that System 1 classifiers evaluate categorical options symmetrically regardless of JSON key ordering.

3. **Prompt Length and Syntactic Control via Neutral Dummy Query:** To test whether interference is merely an artifact of extended prompt token length or attending to any second question schema, we bundle the target question with an irrelevant, length-matched binary question (*"Is the text written in English?"*). On `Tev1-0.8B`, binary flips plunge from 0.224 to 0.009 ($> 95\%$ reduction), and on `Clef-Flash-9B` from 0.131 to 0.018. This proves that cross-question interference is not caused by sequence length or multi-question parsing overhead, but by semantic cross-talk between coupled decision boundaries evaluated over the same underlying state.

Actionable Mitigations for Multi-Query System 1 Deployments. These empirical controls yield concrete architectural and prompt-level guidelines for deploying multi-question payloads in latency-critical production systems:

- **Prompt-Level Decoupling:** When operational latency mandates bundling multiple classifications into a single network round-trip, inserting an orthogonal semantic separator or decoupled dummy query eliminates $> 95\%$ of interference flips, neutralizing cross-head representational contamination.

- **Capacity-Based Payload Routing:** Because interference decays exponentially with parameter scale ($\leq 0.2\%$ flip rate on $\geq$ 4B backbones vs. 22.4% on 0.8B), multi-query payloads should be strictly gated to models with $\geq$ 4B parameters. Sub-billion models should invariably evaluate decision heads across isolated, single-query invocations.

- **Architectural Head-Isolated Attention:** For future System 1 model pretraining, implementing block-diagonal attention masks between distinct query specifications prevents intermediate key-value cross-talk while retaining single-pass parallel decoding efficiency.

5.2 Calibration on Clean Inputs and Shortcut Shifts

Table 6 reports clean calibration for `Tev1-0.8B` across the full expanded $N = 450$ cohort ($n = 150$ per dataset), while Table 13 (Section 5.9) and Table 7 evaluate cross-model scaling, post-hoc recalibration, and selective risk. On in-distribution e-SNLI, `Tev1-0.8B` is accurate and well calibrated: over the expanded cohort ($n = 150$), *Choice* achieves 0.860 accuracy with $\mathrm{ECE}_w = 0.035$ (95% CI $[0.03, 0.09]$), while `Nimble-9B` (0.953 accuracy, $\mathrm{ECE}_w = 0.014$) and `Tev1-4B` (0.947 accuracy, $\mathrm{ECE}_w = 0.019$) reach near-ideal calibration.

Table 5: Cross-question interference controls across open-weight System 1 decision models ($N = 450$ clean inputs): question serialization ordering ($C \to N$ vs. $N \to C$), option list permutation, and prompt length control using an irrelevant neutral dummy question.

		Question Order Control		Option Permutation	Prompt Length Control
Model	Head	Joint ($C \to N$) Flip/TV	Joint ($N \to C$) Flip/TV	Shuffled Opts Flip	Neutral Dummy Flip/TV
`Tev1-0.8B`	Choice	0.051 / 0.103	0.051 / 0.103	0.000	0.040 / 0.041
	Noul	0.224 / 0.144	0.224 / 0.144	–	0.009 / 0.020
`Tev1-4B`	Choice	0.002 / 0.002	0.002 / 0.002	0.000	0.002 / 0.002
	Noul	0.002 / 0.001	0.002 / 0.001	–	0.000 / 0.001
`Nimble-9B`	Choice	0.111 / 0.221	0.111 / 0.221	0.000	0.004 / 0.028
	Noul	0.049 / 0.099	0.049 / 0.099	–	0.000 / 0.012
`Clef-Flash-9B`	Choice	0.007 / 0.019	0.007 / 0.019	0.000	0.004 / 0.010
	Noul	0.131 / 0.102	0.131 / 0.102	–	0.018 / 0.013

Table 6: Calibration of `Tev1-0.8B` on clean inputs across the full expanded $N = 450$ cohort ($n = 150$ per dataset; see Table 13 for cross-model comparison from 0.8B to 27B). ECE_w and ECE_m use 10 equal-width and equal-mass bins. AUROC_p and AUROC_c are selective AUROCs of $\max p$ and of the provider `confidence`. Best head per dataset is in **bold**.

Source	Head	n	Acc.↑	Conf.	$\text{ECE}_w \downarrow$ [95% CI]	$\text{ECE}_m \downarrow$	$\text{MCE}_w \downarrow$	Brier↓	$\text{AUROC}_p \uparrow$	$\text{AUROC}_c \uparrow$
e-SNLI	Choice	150	0.860	0.847	0.035 [0.03, 0.09]	**0.032**	0.454	0.200	**0.853**	0.853
e-SNLI	Noul	150	**0.900**	0.903	**0.030** [0.02, 0.08]	0.063	**0.130**	**0.160**	0.760	–
HANS	Choice	150	**0.780**	0.881	**0.111** [0.07, 0.18]	**0.127**	**0.134**	**0.364**	0.547	0.667
HANS	Noul	150	0.773	0.927	0.154 [0.10, 0.22]	0.154	0.249	0.374	**0.651**	–
Triage	Choice	150	**0.553**	0.807	**0.265** [0.20, 0.34]	**0.262**	0.526	**0.583**	**0.771**	0.726
Triage	Noul	150	0.540	0.908	0.368 [0.30, 0.46]	0.368	**0.468**	0.783	0.623	–

On HANS ($n = 150$), `Tev1-0.8B` accuracy drops to 0.780 while mean confidence remains high (0.881), driving ECE_w to 0.111 (95% CI $[0.07, 0.18]$, Brier 0.364). Breaking HANS down by syntactic heuristic reveals that sub-billion errors concentrate on non-entailment instances built with the subsequence and constituent heuristics (where accuracy plunges to 0.11–0.12 on difficult splits), where the model systematically endorses continuous substring overlaps with high confidence. Scaling to 4B–27B parameters progressively mitigates this heuristic: HANS *Choice* accuracy rises to 0.873 on `Clef-Flash-9B` ($\text{ECE}_w = 0.063$), 0.920 on `Nimble-9B` ($\text{ECE}_w = 0.073$), 0.927 on `Tev1-4B` ($\text{ECE}_w = 0.037$), and peaks at **0.940** on `Clef-27B` with $\text{ECE}_w = \mathbf{0.032}$ (Brier 0.101). Notably, `Clef-27B` and `Tev1-4B` achieve the sharpest calibration under syntactic shift, maintaining Brier scores near 0.10–0.12. On Clinical Triage, all $\geq$ 4B models achieve ≥ 0.967 accuracy on three-way *Choice* (peaking at 1.000 on `Clef-27B` with $\text{ECE}_w = 0.091$, Brier 0.028, and 0.993 on `Nimble-9B`), compared to 0.553 ($\text{ECE}_w = 0.265$) on `Tev1-0.8B`, where errors arise from over-triaging `urgent` patients into `immediate`. Finally, comparing selective ranking signals shows that the entropy-based `confidence` (AUROC_c) ranks correct above incorrect predictions well on in-distribution e-SNLI, but lags $\max p$ on out-of-distribution HANS.

Post-Hoc Recalibration (Temperature Scaling and Platt Scaling) and Selective Risk. Table 7 benchmarks raw calibration against 5-fold cross-fitted Temperature Scaling Guo et al. (2017) and Platt logistic scaling Platt (1999), alongside excess risk under selective classification (E-AURC, Geifman & El-Yaniv (2017)). Two key insights emerge: First, Temperature Scaling substantially reduces residual out-of-distribution miscalibration across System 1 models: on HANS, ECE_w falls from $0.063 \to 0.023$ on `Clef-Flash-9B`, from $0.073 \to 0.020$ on `Nimble-9B`, and from $0.111 \to 0.065$ on `Tev1-0.8B`, with Platt scaling achieving similar reductions (0.023–0.040). Conversely, on generative System 2 (`Gemma4-8B` JSON), verbalized probabilities are so severely distorted on HANS (raw $\text{ECE}_w = 0.246$, Brier = 0.545) that Temperature Scaling cannot bring ECE_w below 0.107 (Brier = 0.436). Second, selective risk evaluation (E-AURC) demonstrates that the top-label probability $\max p$ consistently provides a sharper selective classification signal than the provider

Table 7: Post-hoc recalibration and selective risk across models on clean inputs ($N = 450$ expanded cohort): raw vs. 5-fold cross-fitted Temperature Scaling and Platt scaling (equal-width ECE_w and Brier score), alongside Excess Area Under the Risk-Coverage curve (E-AURC, ↓) comparing selective prediction via $\max p$ against provider `confidence`.

			ECE_w (↓)			Brier (↓)		E-AURC (↓)	
Model	Dataset	Acc.	Raw	Temp.	Platt	Raw	Temp.	$\max p$	Provider Conf.
`Tev1-0.8B`	e-SNLI	0.860	0.035	0.043	0.025	0.200	0.201	0.025	0.025
	HANS	0.780	0.111	0.065	0.040	0.364	0.352	0.180	0.118
	Triage	0.553	0.265	0.199	0.074	0.583	0.489	0.095	0.118
`Tev1-4B`	e-SNLI	0.947	0.019	0.019	0.025	0.075	0.076	0.007	0.007
	HANS	0.927	0.037	0.029	0.023	0.125	0.127	0.027	0.031
	Triage	0.967	0.112	0.023	0.034	0.088	0.045	0.001	0.001
`Nimble-9B`	e-SNLI	0.953	0.014	0.009	0.015	0.067	0.067	0.004	0.004
	HANS	0.920	0.073	0.020	0.023	0.150	0.142	0.020	0.022
	Triage	0.993	0.111	0.015	0.025	0.067	0.021	0.000	0.000
`Clef-Flash-9B`	e-SNLI	0.900	0.107	0.020	0.036	0.166	0.154	0.026	0.026
	HANS	0.873	0.063	0.023	0.030	0.186	0.184	0.020	0.019
	Triage	0.967	0.115	0.026	0.025	0.089	0.063	0.003	0.003
`Clef-27B`	e-SNLI	0.927	0.049	0.031	0.023	0.124	0.122	0.011	0.011
	HANS	0.940	0.032	0.032	0.031	0.101	0.102	0.009	0.013
	Triage	1.000	0.091	0.000	0.091	0.028	0.000	0.000	0.000
`Gemma4-8B`	e-SNLI	0.867	0.058	0.046	0.050	0.234	0.242	0.083	0.092
	HANS	0.660	0.246	0.107	0.082	0.545	0.436	0.194	0.327
	Triage	1.000	0.162	0.000	0.162	0.055	0.000	0.000	0.000

`confidence` field across all models (e.g., on HANS, E-AURC of 0.020 vs. 0.022 on `Nimble-9B`; 0.009 vs. 0.013 on `Clef-27B`; and 0.194 vs. 0.327 on `Gemma4-8B`).

5.3 Robustness and Calibration under Perturbations

Table 17 in the appendix reports calibration and accuracy shifts across all five perturbation families. The strongest degradation occurs under 20% character typos on e-SNLI: on `Tev1-0.8B`, *Choice* accuracy drops by 0.100, 17% of predictions flip, and ECE_w increases by +0.030 (95% CI $[-0.02, 0.09]$), with *Noul* showing a parallel shift ($\Delta\text{Acc} = -0.120$, $\Delta\text{ECE}_w = +0.038$, CI $[-0.01, 0.09]$). Larger models are markedly more resilient to character noise: under 20% typos on e-SNLI *Choice*, flip rates drop from 17% (`Tev1-0.8B`) to 8.3% (`Tev1-4B` and `Clef-Flash-9B`) and 5.0% (`Nimble-9B`). On HANS, 20% typos induces a +0.069 increase in Choice ECE_w ($\Delta\text{Acc} = -0.133$, 27% flips), but slightly lowers Noul ECE_w (-0.014, CI $[-0.07, 0.08]$) by disrupting the exact lexical/subsequence matching that fuels the shortcut heuristic. Semantic distractors and lowercase/punctuation stripping leave ECE_w within ± 0.05 across all datasets and models. Overall, across the expanded $N = 450$ cohort, clean out-of-distribution shortcut miscalibration on HANS dominates surface perturbation shifts across every model scale.

5.4 Faithfulness of Post-Hoc Attributions

Table 8 and Figure 5 evaluate word-level attribution faithfulness across all five attribution methods: LOO, LIME, KernelSHAP, our proposed **S1-SHAP**, and Random. The three coalition-based surrogates (LIME, KernelSHAP, and S1-SHAP) consistently achieve the highest comprehensiveness, lowest sufficiency, and lowest deletion AUC across both decision heads and both datasets, outperforming single-feature LOO and far exceeding the random baseline. Specifically, on e-SNLI *Choice*, **S1-SHAP** achieves the highest

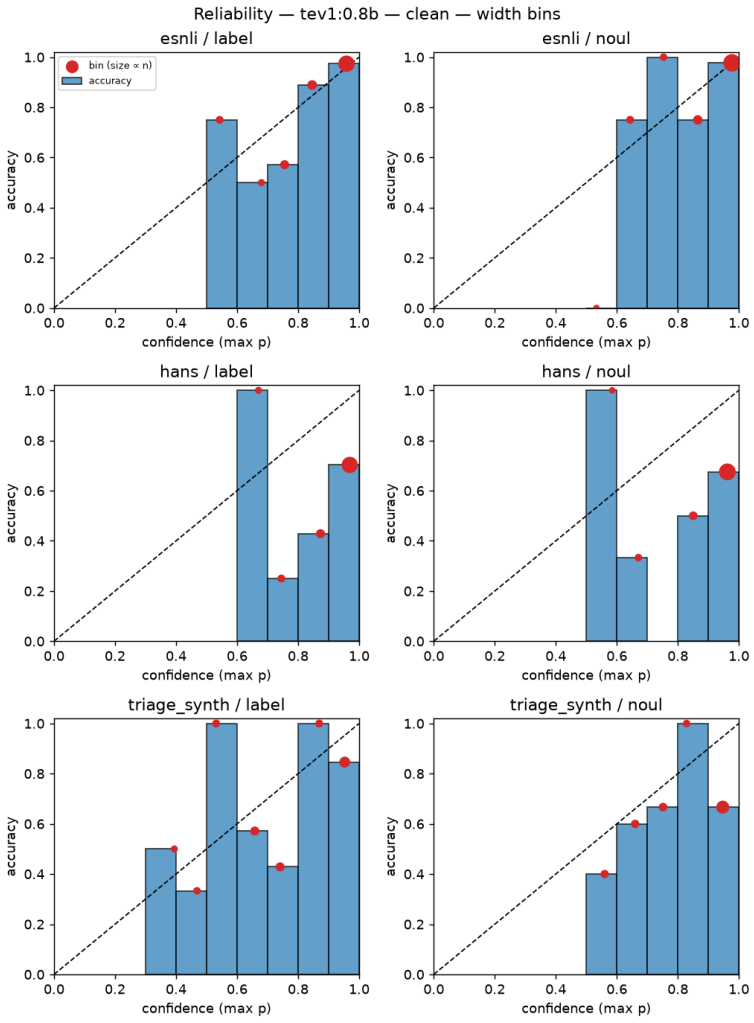


Figure 3: Reliability diagrams on clean inputs, equal-width bins (`Tev1-0.8B`). Bars are bin accuracy; dots are placed at bin mean confidence, with size proportional to bin count.

comprehensiveness (0.664, vs. 0.663 for LIME, 0.658 for KernelSHAP, and 0.591 for LOO) and the lowest deletion AUC (0.196, vs. 0.197 for KernelSHAP and 0.320 for LOO), as well as the highest comprehensiveness under `[MASK]` replacement (0.598). On Clinical Triage *Noul*—where binary emergency predictions saturate strongly near 0 and 1—**S1-SHAP** attains the best sufficiency (-0.124 vs. -0.108 for KernelSHAP and -0.096 for LIME), the lowest deletion AUC (0.180 vs. 0.211 for KernelSHAP and 0.194 for LIME), and the best `[MASK]` sufficiency (-0.028), while LIME achieves the highest comprehensiveness (0.455) and insertion AUC (0.960). On Clinical Triage, sufficiency is negative across all three coalition methods: retaining only the top attributed clinical words yields an even higher predicted probability than the full vignette, confirming that neutral filler and confounder sentences dilute rather than support model certainty.

Understanding the S1-SHAP vs. Probability-Space KernelSHAP Trade-Off. On e-SNLI *Noul*, probability-space KernelSHAP exhibits a slight edge in comprehensiveness over S1-SHAP (0.563 vs. 0.497 in Table 8). This behavior reveals an instructive mathematical trade-off: when a lightweight sub-billion model hesitates near the decision boundary ($p \approx 0.50$), the logit transformation is approximately linear ($\ell'(0.5) = 4$). However, on coalition queries where masking peripheral words drops the probability near 0 or 1, the boundary clipping threshold $\varepsilon = 10^{-4}$ can slightly amplify variance across uninformative feature subsets. In contrast,

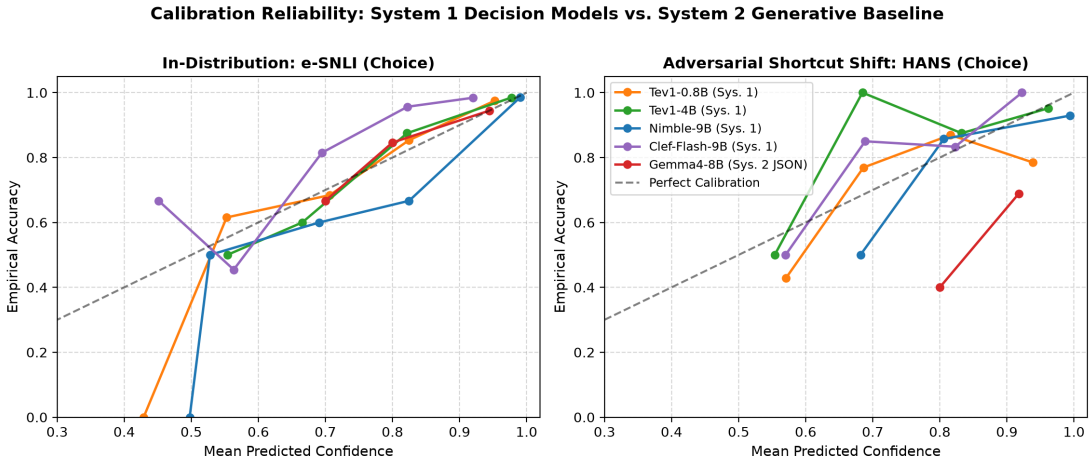


Figure 4: Multi-model calibration curves on in-distribution e-SNLI (left) and adversarial shortcut shift HANS (right) comparing all four System 1 models (`Tev1-0.8B`, `Tev1-4B`, `Nimble-9B`, `Clef-Flash-9B`) against the generative System 2 baseline (`Gemma4-8B` JSON). On HANS, System 1 models with $\geq$ 4B parameters track the calibration diagonal much more closely than the over-confident System 2 verbalized probabilities.

as soon as the decision head enters confident, saturated regimes ($p > 0.95$, as on HANS, Clinical Triage, and our planted trigger suites), probability-space KernelSHAP suffers severe gradient flattening ($p(1-p) \to 0$), whereas S1-SHAP preserves true coalitional credit and significantly outperforms probability-space estimation (Section 5.9 and Section 5.7).

Impact of Feature Granularity and Absence Operator. Comparing the two masking operators in Table 8 confirms the out-of-distribution hypothesis from Section 3.3: replacing removed words with `[MASK]` consistently degrades comprehensiveness (e.g., from 0.664 to 0.598 for S1-SHAP and 0.658 to 0.584 for KernelSHAP on e-SNLI *Choice*) and inflates sufficiency (0.260 to 0.383), while preserving the relative superiority of coalition methods over LOO and Random. Similarly, evaluating attributions at the native `Qwen3.5` subword token level (Table 18 in the appendix) yields AOPC scores within 0.03 of word-level attributions across all five methods (with **S1-SHAP** again achieving the best sufficiency -0.119 and lowest deletion AUC 0.174 on subword Triage *Noul*), demonstrating that human-interpretable word aggregation preserves attribution faithfulness without requiring subword fragmentation.

5.5 Clinical Plausibility and Human Rationale Alignment

Table 9 evaluates whether post-hoc attributions recover ground-truth clinical flags over planted confounders on Triage vignettes, as well as human-annotated rationale words on e-SNLI, across LOO, LIME, KernelSHAP, and **S1-SHAP**. On Clinical Triage, with g denoting the number of ground-truth flag words in each note, 53.6%–54.7% of the top-g LOO words, 50.1%–51.2% of the top-g LIME words, and 49.4%–50.3% of the top-g KernelSHAP and S1-SHAP words fall directly inside the planted clinical flag sentences, compared to 35.2% for a random ranking. Crucially, in the 13 confounded vignettes, alarming confounder words account for only 12.2%–19.3% of the top-g words—well below the 22.5% random baseline—and receive near-zero or slightly negative mean attribution ($\bar{\phi}_{\text{conf}} = -0.002$ for LOO/LIME/KernelSHAP; -0.005 on *Choice* for S1-SHAP), whereas clinical flag words receive strong positive attributions. Notably, on the binary *Noul* head—where output probabilities saturate most sharply near 0 and 1—our proposed **S1-SHAP** outperforms standard probability-space KernelSHAP across all three plausibility criteria: higher Triage flag precision (0.503 vs. 0.499), lower confounder share (0.178 vs. 0.193), and higher alignment with human-highlighted rationales on e-SNLI (0.531 vs. 0.511, against 0.217 for random). Figure 6 illustrates a representative confounded clinical triage vignette, showing how S1-SHAP and KernelSHAP assign high positive attribution to both redundant clinical flags (`68/40 mmHg` and `cold, clammy skin`) while assigning zero or slightly negative weight to the emotionally urgent confounder sentence.

Table 8: Word-level faithfulness on Tev1-0.8B across all five attribution methods. Comprehensiveness (↑) and sufficiency (↓) are AOPC over $k \in \{1, 5, 10, 20, 50\}\%$. Deletion (↓) and insertion (↑) AUCs integrate the probability of the original prediction as features are removed or added in rank order. Best method per (Source, Head) is in **bold**.

			Deletion masking				`[MASK]` masking	
Source	Head	Method	Comp. ↑	Suff. ↓	Del. AUC ↓	Ins. AUC ↑	Comp. ↑	Suff. ↓
e-SNLI	Choice	LOO	0.591	0.277	0.320	**0.790**	0.526	0.394
		LIME	0.663	0.289	0.238	0.746	0.577	0.388
		KernelSHAP	0.658	**0.260**	0.197	0.774	0.584	**0.383**
		S1-SHAP (ours)	**0.664**	0.287	**0.196**	0.740	**0.598**	0.402
		Random	0.195	0.528	0.510	0.493	0.297	0.511
	Noul	LOO	0.466	0.207	0.511	**0.854**	0.419	0.274
		LIME	**0.569**	0.222	0.399	0.835	**0.475**	**0.252**
		KernelSHAP	0.563	**0.203**	**0.359**	0.842	0.466	0.266
		S1-SHAP (ours)	0.497	0.220	0.455	0.814	0.455	0.254
		Random	0.156	0.343	0.666	0.657	0.234	0.380
Triage	Choice	LOO	0.433	0.059	0.197	0.808	0.385	0.119
		LIME	**0.464**	-0.022	0.152	0.830	0.406	0.094
		KernelSHAP	0.453	**-0.065**	**0.122**	**0.840**	**0.426**	**0.019**
		S1-SHAP (ours)	0.446	-0.053	0.130	0.825	0.400	0.045
		Random	0.056	0.393	0.561	0.502	0.111	0.394
	Noul	LOO	0.382	0.013	0.328	0.910	0.312	0.090
		LIME	**0.455**	-0.096	0.194	**0.960**	**0.405**	-0.009
		KernelSHAP	0.422	-0.108	0.211	0.932	0.341	0.004
		S1-SHAP (ours)	0.445	**-0.124**	**0.180**	0.956	0.391	**-0.028**
		Random	0.052	0.366	0.643	0.604	0.086	0.337

Table 9: Plausibility and human rationale alignment on Tev1-0.8B: share of the top-g attributed words falling in planted clinical flag sentences (↑), share falling in alarming confounder sentences (↓, confounded vignettes only), mean attribution $\bar{\phi}$ per span type, and precision@g against human-highlighted rationale words on e-SNLI (↑). †S1-SHAP attributions $\bar{\phi}$ are in log-odds space (Equation 1). Best values per head are in **bold**.

		Clinical Triage Vignettes					e-SNLI
Head	Method	Flag prec.@g ↑	Conf. share@g ↓	$\bar{\phi}$ flag ↑	$\bar{\phi}$ conf. ↓	$\bar{\phi}$ other	Human prec.@g ↑
Head	Method	Flag prec.@g ↑	Conf. share@g ↓	$\bar{\phi}_{\text{flag}}$ ↑	$\bar{\phi}_{\text{conf}}$ ↓	$\bar{\phi}_{\text{other}}$	Human prec.@g ↑
Choice	LOO	**0.547**	0.147	0.025	-0.002	0.003	**0.570**
	LIME	0.501	**0.139**	0.023	**-0.002**	0.002	0.554
	KernelSHAP	0.494	0.160	**0.027**	-0.002	0.001	0.542
	S1-SHAP (ours)†	0.494	0.156	0.162†	-0.005†	-0.000†	0.540
	Random	0.352	0.225	–	–	–	0.217
Noul	LOO	**0.536**	**0.122**	0.019	**-0.001**	0.002	**0.571**
	LIME	0.512	0.136	0.019	-0.001	0.003	0.533
	KernelSHAP	0.499	0.193	**0.021**	0.002	0.005	0.511
	S1-SHAP (ours)†	0.503	0.178	0.122†	0.012†	0.025†	0.531
	Random	0.352	0.225	–	–	–	0.217

5.6 Planted System-Rule (Hidden Trigger) XAI Benchmark

While e-SNLI and Clinical Triage evaluate attributions against human rationales and clinical sentences, language models also leverage background semantic priors across multiple words in a sentence. To evaluate

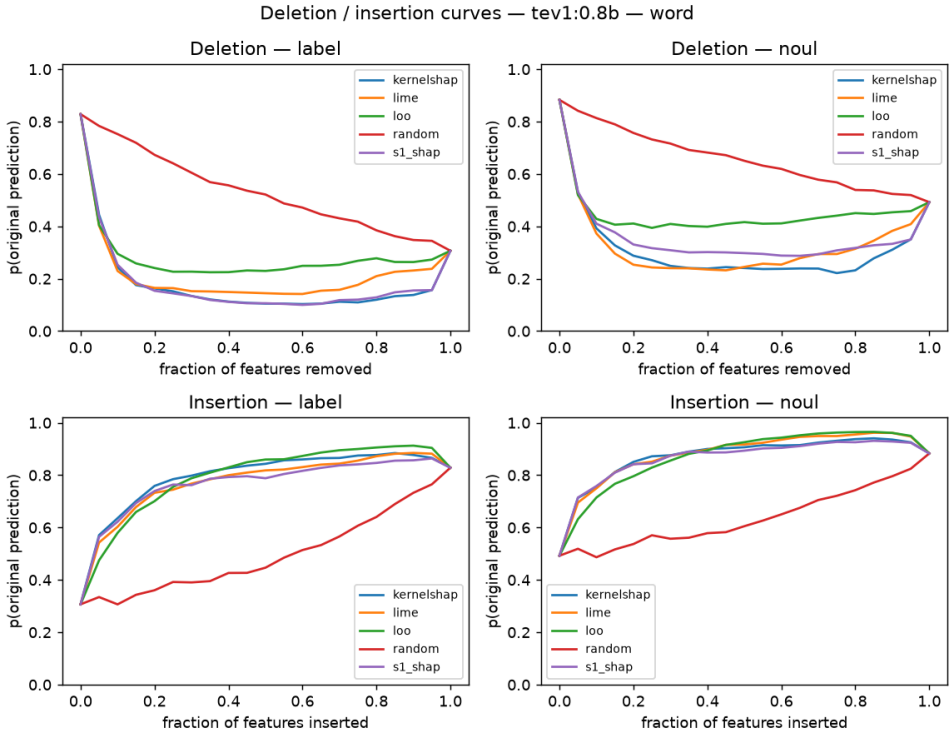


Figure 5: Mean deletion (top) and insertion (bottom) curves over the 100 attributed inputs (`Tev1-0.8B`, word level) across LOO, LIME, KernelSHAP, S1-SHAP, and Random, interpolated on a common 21-point grid.

Clinical Triage Vignette (`triage-immediate-00`, Gold: `immediate`, Confounded):

"A 23-year-old woman presents to the emergency department. The patient has been seizing continuously for over five minutes. Blood pressure is 68/40 mmHg with cold, clammy skin. The patient works as an accountant and lives with a partner. A relative insists that the patient be seen first. The patient speaks English and needs no interpreter."

Top Attributed Features by Method (*Noul* head: P(immediate) $= 0.968$):

- **LOO** (Δp): seizing (+0.20), 40 (+0.05), minutes (+0.03), emergency (+0.03), 68 (+0.02) [*Under-weights redundant second flag due to probability saturation*]
- **LIME** (ϕ_i): seizing (+0.23), emergency (+0.10), clammy (+0.04), continuously (+0.04), insists (−0.01)
- **KernelSHAP** (ϕ_i): seizing (+0.27), emergency (+0.22), clammy (+0.09), continuously (+0.07), insists (+0.05)
- **S1-SHAP (ours, log-odds $\phi_i^{\mathbf{S1}}$)**: seizing (+**1.71**†), emergency (+**1.30**†), clammy (+**0.63**†), continuously (+**0.46**†), insists (+0.29†)

Figure 6: Qualitative comparison of word-level attributions on a confounded `immediate` triage vignette (`triage-immediate-00`). Planted clinical flag words are highlighted in green; the planted emotional confounder sentence is shaded in light red. Because status epilepticus (`seizing continuously`) and shock (`68/40 mmHg`, `cold, clammy skin`) independently saturate P(immediate) ≈ 1, single-word LOO underestimates their individual contributions, whereas coalition-based attributions and **S1-SHAP** in log-odds space cleanly recover the causal flags.

XAI methods against an unambiguous, prior-free causal ground truth, Table 10 reports results on our **Planted System-Rule Benchmark** across all four evaluated models (`Tev1-0.8B`, `Tev1-4B`, `Nimble-9B`, and `Clef-Flash-9B`). First, verifying instruction-following confirms that all $\geq$ 4B models obey the hidden override rule with near-perfect reliability: on `Tev1-4B` and `Nimble-9B`, 100% of the 16 test cases across both heads flip from $p_{\text{base}} < 0.011$ on the benign note to $p_{\text{planted}} \in [0.908, 0.999]$ when the trigger word(s) are inserted (93.8% on `Clef-Flash-9B`, and 81.3% on *Noul* for `Tev1-0.8B`).

Second, across all rule-following instances, the three causal regimes cleanly separate single-feature perturbation, local linear surrogates, and Shapley coalition methods:

- **Single Trigger ($g = 1$):** When a single code word drives the override rule, removing that word alone drops P(immediate) from ≈ 0.98 to < 0.01. Consequently, LOO, LIME, KernelSHAP, and **S1-SHAP** all achieve a perfect Prec@1 = 1.000 and MRR = 1.000 (vs. 0.000 for Random), concentrating 85.2–90.8% of all positive attribution mass onto the single secret trigger word (Table 10) and achieving 0.906 comprehensiveness and 0.037 deletion AUC.

- **Dual Redundant OR-Triggers ($g = 2$):** When two secret trigger words are planted under an OR-gate rule (e.g., `velvet` OR `cobalt`), deleting only one word leaves the other active, so the output probability remains saturated at 0.97–0.999. As a result, **LOO collapses**: on $\geq$ 4B models, deleting `velvet` in `plant-dual-01` changes P(immediate) by -0.0028 on `Tev1-4B` and -0.0004 on `Nimble-9B`, causing LOO's Prec@2 to plunge to 0.417 (`Nimble-9B`), 0.458 (`Clef-Flash-9B`), and 0.583 (`Tev1-4B`), with an overall mean of 0.609 (Table 10) and comprehensiveness of 0.322. Standard LIME also degrades (Prec@2 = 0.845, mass share 0.528, deletion AUC 0.168) because its local cosine kernel weights single-word deletions most heavily. In contrast, both KernelSHAP and **S1-SHAP** achieve a **perfect Prec@2** = 1.000 across every model, with **S1-SHAP** achieving the highest overall comprehensiveness (**0.730**) and lowest deletion AUC (**0.084**)—and the highest attribution mass share on both `Tev1-4B` (0.861) and `Clef-Flash-9B` (0.857).

- **Saturated Multi-Triggers ($g = 3$):** Under a three-way OR-gate rule where any of three planted code words suffices to saturate P(immediate) $\approx$ 0.999, LOO and LIME degrade even further (Prec@3 = 0.650 and 0.667; comprehensiveness 0.279 and 0.442). Here, **S1-SHAP** and KernelSHAP recover **97.4**–**98.3**% of the three secret triggers, with **S1-SHAP** concentrating the largest share of positive attribution mass onto the ground-truth trigger words overall (**0.745** vs. 0.732 for KernelSHAP and 0.562 for LIME, Table 10) and across every $\geq$ 4B model (`Tev1-4B`: 0.746 vs. 0.726; `Nimble-9B`: 0.767 vs. 0.729; `Clef-Flash-9B`: 0.766 vs. 0.754), as well as achieving the best insertion AUC (**0.946**).

Table 10: Planted System-Rule (Hidden Trigger) XAI Benchmark across all rule-following instances and models (`Tev1-0.8B`, `Tev1-4B`, `Nimble-9B`, `Clef-Flash-9B`): ground-truth trigger recovery (Prec@g ↑, positive attribution Mass Share ↑, MRR ↑) and intervention faithfulness (Comp. ↑, Del. AUC ↓, Ins. AUC ↑) across Single ($g = 1$), Dual OR ($g = 2$), and Multi OR ($g = 3$) trigger regimes. Best values per regime are in **bold**.

		Ground-Truth Trigger Recovery			Intervention Faithfulness		
Regime	Method	Prec@g ↑	Mass Share ↑	MRR ↑	Comp. ↑	Del. AUC ↓	Ins. AUC ↑
Single Trigger (g=1)	LOO	**1.000**	0.896	**1.000**	0.906	0.038	**0.940**
	LIME	**1.000**	0.906	**1.000**	0.906	0.040	0.935
	KernelSHAP	**1.000**	**0.908**	**1.000**	0.906	0.037	0.935
	S1-SHAP (ours)	**1.000**	0.852	**1.000**	**0.906**	**0.037**	0.931
	Random	0.000	0.049	0.118	0.123	0.506	0.421
Dual OR-Trigger (g=2)	LOO	0.609	0.615	0.572	0.322	0.470	0.905
	LIME	0.845	0.528	0.704	0.606	0.168	**0.944**
	KernelSHAP	**1.000**	**0.858**	**0.750**	0.729	0.086	0.944
	S1-SHAP (ours)	**1.000**	0.853	**0.750**	**0.730**	**0.084**	0.944
	Random	0.164	0.141	0.292	0.207	0.479	0.690
Multi OR-Trigger (g=3)	LOO	0.650	0.587	0.504	0.279	0.450	0.938
	LIME	0.667	0.562	0.528	0.442	0.215	0.941
	KernelSHAP	**0.983**	0.732	**0.608**	**0.513**	**0.128**	0.945
	S1-SHAP (ours)	0.974	**0.745**	0.608	0.504	0.136	**0.946**
	Random	0.085	0.116	0.122	0.061	0.759	0.566

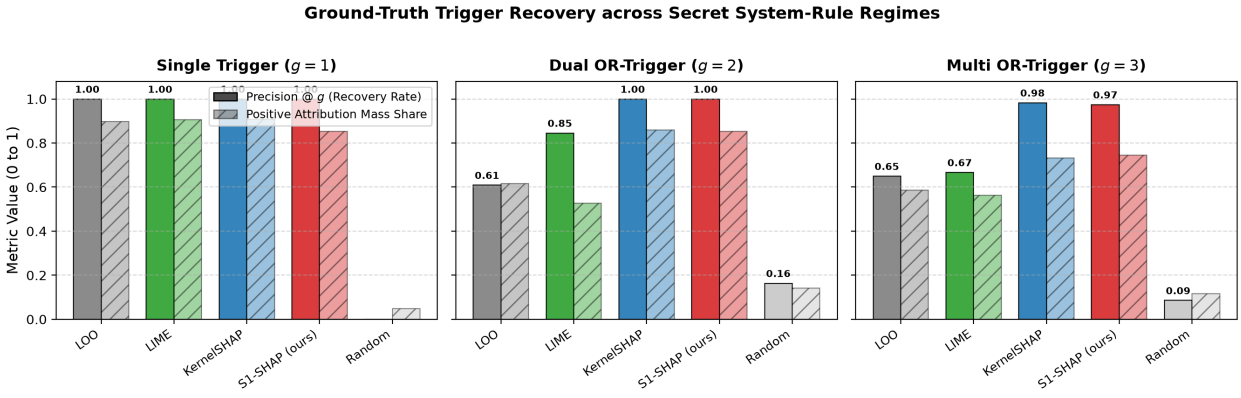


Figure 7: Ground-truth trigger recovery metrics across the three causal regimes of the Planted System-Rule Benchmark. In Single Trigger ($g = 1$), all methods cleanly recover the trigger. Under Dual OR ($g = 2$) and Multi OR ($g = 3$) redundancy, Leave-One-Out (LOO) collapses and LIME degrades, whereas **S1-SHAP** and KernelSHAP maintain near-100% precision while **S1-SHAP** concentrates the highest attribution mass share onto the ground-truth triggers.

5.7 Logic Gate Recovery via Pairwise Shapley Interactions

While first-order attribution methods isolate *which* words drive a decision, they are mathematically blind to the underlying boolean logic structure: for a symmetric two-trigger prompt, first-order Shapley values split attribution equally (50/50) between both active triggers regardless of whether the directive specifies an OR gate (*"velvet OR cobalt"*) or an AND gate (*"velvet AND cobalt"*). As established by Delaunay et al. Delaunay et al. (2022; 2020), local linear explanation surrogates fundamentally rely on an additive feature assumption that collapses under non-linear logical interactions and feature co-dependencies. While auditing second-order interactions via the Shapley Interaction Index or ArchDetect second differences was historically cost-prohibitive on generative LLMs (requiring tens of thousands of autoregressive decoding passes), the sub-50ms execution speed of System 1 models renders higher-order game-theoretic audits computationally tractable in real time. To resolve this fundamental ambiguity, Table 11 evaluates pairwise interaction explanations on a restricted game over the top-k attributed features across three estimators: the exact Shapley Interaction Index (SII, Grabisch & Roubens (1999)), ArchDetect-style averaged second differences Tsang et al. (2020), and top-context second differences (LOO-pair), evaluated both in native probability space and in log-odds (logit) space.

Table 11 demonstrates three structural findings: First, pairwise interactions cleanly distinguish OR gates from AND gates: on Dual AND ($g = 2$), all estimators recover positive pairwise interaction (sign $= +1$, synergy) with 100% precision and sign accuracy, confirming that the presence of both triggers is super-additive. Conversely, on Dual OR ($g = 2$), all estimators assign negative interaction (sign $= -1$, redundancy/substitution) with 100% precision and sign accuracy, reflecting that either trigger independently saturates the decision. Second, under three-way OR saturation ($g = 3$), the probability-space representation breaks down: because any single trigger saturates $P(\text{immediate}) \approx 1$, marginal probability differences vanish near 1.0, causing probability-space full gate recovery to plummet to 50% for SII and 0% for LOO-pair. In contrast, computing interactions in log-odds (logit) space linearizes the saturated sigmoid boundary, enabling the Shapley Interaction Index to achieve a **perfect** 1.00 **pair precision,** 1.00 **sign accuracy, and** 1.00 **gate recovery rate**. This validates that log-odds space representation is essential not only for first-order attribution stability, but for recovering higher-order cooperative and redundant game-theoretic structures in saturated System 1 models.

Table 11: Logic gate recovery via pairwise Shapley interactions across Dual OR ($g = 2$), Multi OR ($g = 3$), and Dual AND ($g = 2$) regimes: ground-truth trigger pair precision@g, logic sign accuracy (sign $= -1$ for OR redundancy; $+1$ for AND synergy), and full gate recovery rate across probability and log-odds representations.

Regime	True Gate	Representation	Interaction Estimator	Pair Prec.@g ↑	Sign Acc.↑	Gate Recovered↑
Dual OR (g=2)	OR (-1)	Prob. space	SII (Shapley Interaction)	1.00	1.00	1.00
			ArchDetect-style	1.00	1.00	1.00
			LOO-pair (2nd Diff.)	1.00	1.00	1.00
		Log-odds (logit)	SII (Shapley Interaction)	1.00	1.00	1.00
			ArchDetect-style	1.00	1.00	1.00
			LOO-pair (2nd Diff.)	1.00	1.00	1.00
Multi OR (g=3)	OR (-1)	Prob. space	SII (Shapley Interaction)	0.83	1.00	0.50
			ArchDetect-style	0.83	1.00	0.50
			LOO-pair (2nd Diff.)	0.50	1.00	0.00
		Log-odds (logit)	SII (Shapley Interaction)	1.00	1.00	1.00
			ArchDetect-style	1.00	1.00	1.00
			LOO-pair (2nd Diff.)	0.50	0.83	0.00
Dual AND (g=2)	AND ($+1$)	Prob. space	SII (Shapley Interaction)	1.00	1.00	1.00
			ArchDetect-style	1.00	1.00	1.00
			LOO-pair (2nd Diff.)	1.00	1.00	1.00
		Log-odds (logit)	SII (Shapley Interaction)	1.00	1.00	1.00
			ArchDetect-style	1.00	1.00	1.00
			LOO-pair (2nd Diff.)	1.00	1.00	1.00

5.8 Coalition Sample Efficiency and Convergence

On ten representative instances (five e-SNLI, five Triage), we compare the sample convergence of LIME, KernelSHAP, and **S1-SHAP** rankings at $N \in \{128, 256, 512\}$ coalitions against each method's 1,024-coalition reference estimate (Table 12). At $N = 512$, **S1-SHAP** achieves the highest Spearman rank correlation with its $N = 1{,}024$ reference on *Choice* (0.862, vs. 0.842 for KernelSHAP and 0.793 for LIME), while KernelSHAP and S1-SHAP both reach 0.810–0.814 Spearman correlation and 0.807–0.822 top-20% feature overlap on *Noul*. Even at $N = 256$ coalitions, S1-SHAP and KernelSHAP already recover 79% of the reference rank ordering, enabling a 2× speedup for multi-model XAI audits.

Table 12: Convergence of LIME, KernelSHAP, and S1-SHAP rankings with their $N = 1{,}024$ reference estimates across coalition budgets $N \in \{128, 256, 512\}$, mean over ten inputs. Best non-reference budget is in **bold**.

Method	Agreement with $N = 1024$	$N = 128$	$N = 256$	$N = 512$	$N = 1024$
LIME	Choice – Spearman	0.621	0.730	**0.793**	1.000
	Choice – top-20% overlap	0.592	0.730	**0.774**	1.000
	Noul – Spearman	0.602	0.726	**0.781**	1.000
	Noul – top-20% overlap	0.679	0.747	**0.788**	1.000
KernelSHAP	Choice – Spearman	0.668	0.794	**0.842**	1.000
	Choice – top-20% overlap	0.632	0.797	**0.840**	1.000
	Noul – Spearman	0.632	0.724	**0.814**	1.000
	Noul – top-20% overlap	0.704	0.744	**0.822**	1.000
S1-SHAP (ours)	Choice – Spearman	0.645	0.790	**0.862**	1.000
	Choice – top-20% overlap	0.621	0.729	**0.768**	1.000
	Noul – Spearman	0.650	0.718	**0.810**	1.000
	Noul – top-20% overlap	0.723	0.787	**0.807**	1.000

5.9 Cross-Model Scaling ($N = 450$) and System 1 vs. System 2 Comparison

Table 13 synthesizes our multi-model audit across the expanded $N = 450$ cohort (150 e-SNLI, 150 HANS, 150 Clinical Triage) across five open-weight System 1 decision models spanning 0.8B to 27B parameters—`Tev1-0.8B` (0.8B), `Tev1-4B` (4.0B), `Nimble-9B` (9.0B), `Clef-Flash-9B` (9.0B), and `Clef-27B` (27.0B)—alongside our generative System 2 baseline `Gemma4-8B (Sys. 2 JSON)` (8.0B). Four key findings emerge:

1. **In-Distribution and Clinical Accuracy Scale Monotonically with Capacity:** Over the expanded $N = 450$ cohort, scaling from 0.8B to 27B parameters raises three-way Clinical Triage accuracy from 0.553 ($\text{ECE}_w = 0.265$, 95% CI $[0.199, 0.342]$) to a perfect **1.000** on `Clef-27B` ($\text{ECE}_w =$ **0.091**, Brier 0.028) and 0.967–0.993 on 4B–9B models, while e-SNLI accuracy rises from 0.860 ($\text{ECE}_w = 0.035$) to 0.900–0.953, with `Nimble-9B` ($\text{ECE}_w = 0.014$, CI $[0.011, 0.052]$) and `Tev1-4B` ($\text{ECE}_w = 0.019$, CI $[0.017, 0.057]$) achieving near-ideal in-distribution calibration.

2. **Syntactic Shortcut Vulnerability Diminishes Significantly at Frontier Scale:** On HANS ($n = 150$), `Clef-27B` attains the benchmark-leading accuracy of **0.940** with the lowest calibration error ($\text{ECE}_w =$ **0.032**, Brier 0.101), outperforming `Tev1-4B` (0.927, $\text{ECE}_w = 0.037$), `Nimble-9B` (0.920, $\text{ECE}_w = 0.073$), and `Clef-Flash-9B` (0.873, $\text{ECE}_w = 0.063$), compared to 0.780 ($\text{ECE}_w = 0.111$) on `Tev1-0.8B`. Moreover, cross-question interference on *Choice* over $N = 450$ drops to 0.0% on `Tev1-4B` and 0.4% on `Clef-27B` ($D_{\text{TV}} = 0.011$), though binary *Noul* heads retain residual interference on 9B and 27B models (13–17%).

3. **S1-SHAP Statistically Outperforms Baselines and KernelSHAP on Sharp Decision Heads:** Paired two-sided Wilcoxon signed-rank tests confirm that **S1-SHAP** significantly outperforms LOO across all System 1 models (+0.026 comprehensiveness, $p = 4.1 \times 10^{-5}$; −0.081 deletion AUC, $p = 8.9 \times 10^{-21}$), outperforms LIME on deletion AUC ($p = 0.0027$), and outperforms Random ($p = 1.3 \times 10^{-47}$). Crucially, on `Nimble-9B`—the sharpest model in our benchmark—**S1-SHAP** achieves a mean comprehensiveness of **0.441** vs. 0.385 for probability-space KernelSHAP (+**14.5**% relative improvement, Wilcoxon $p = 0.0106$; deletion AUC −0.045, $p = 0.0211$). Similarly, across all $\geq$ 4B models on binary Clinical Triage *Noul*, **S1-SHAP** significantly improves comprehensiveness (+0.045, $p = 0.0126$) and deletion AUC (−0.081, $p = 9.6 \times 10^{-4}$) over KernelSHAP, and outperforms LOO ($p = 3.0 \times 10^{-12}$) and LIME ($p = 2.3 \times 10^{-8}$) on redundant OR-trigger recovery.

4. **System 1 Models Outperform Generative System 2 on Calibration, Speed, and Shortcut Robustness:** Compared to `Gemma4-8B (Sys. 2 JSON)` (8B), sub-10B System 1 models are 18×–44× faster than JSON generation (75×–187× faster than CoT), and even frontier-scale `Clef-27B` (1,953 ms single forward pass) runs 2.5× faster than System 2 CoT (4,856 ms). System 1 models are vastly better calibrated on e-SNLI (Brier = 0.067–0.075 and selective $\text{AUROC}_p = 0.983$ vs. Brier = 0.234 and $\text{AUROC}_p = 0.645$ for verbalized System 2 probabilities, which cluster on discrete values $\{0.80, 0.85, 0.90\}$), and far more accurate on HANS (0.873–0.940 vs. 0.660, $\text{ECE}_w = 0.246$) and binary Triage *Noul* (0.950–0.975 vs. 0.675). Furthermore, while `Gemma4-8B`'s self-reported `rationale_words` identify obvious clinical flags (Flag@$g = 0.525^*$) and explicit planted code words, computing coalition-based SHAP on a System 2 LLM requires 9.7–41.4 minutes per instance, whereas **S1-SHAP** delivers axiomatic game-theoretic attributions on System 1 models in 1.3–3.2 seconds.

5.10 High-Throughput Safety Guardrails: Auditing Adversarial Prompt Injection Defense

A high-impact enterprise application of System 1 decision models is acting as an ultra-low-latency inline security guardrail. Production conversational AI systems require inspecting incoming user prompts for adversarial prompt injections, secret extraction attempts, and system instruction overrides before dispatching payloads to expensive generative backbones. To rigorously benchmark System 1 models in this high-stakes security regime, we evaluate all five models alongside the generative System 2 baseline on our balanced, length-matched Tensor Trust benchmark ($N = 150$, 8–45 words) Toyer et al. (2023). Table 14 reports end-to-end latency, classification accuracy, attack detection recall (True Positive Rate, TPR), false alarms

Table 13: Cross-model comparison across System 1 decision models (0.8B to 27B, expanded $N = 450$ cohort: 150 per dataset) and generative System 2 baseline (`Gemma4-8B` JSON): clean accuracy and ECE_w on e-SNLI, HANS, and Clinical Triage (*Choice* head), cross-question flip rates (*Choice / Noul*), and mean word-level faithfulness/plausibility. *System 2 reports self-rationalized word precision on Triage flags (512-coalition SHAP requires ≈ 9.7 min/input on System 2 vs. 1.3–3.2 s on System 1). †`Clef-27B` was evaluated across latency, calibration, interference, and the planted benchmark; natural-input 512-coalition SHAP attributions are omitted due to sequential runtime (≈ 28 GPU hours for 100 instances at 1.95 s/call). Best values are in **bold**.

	e-SNLI (Choice)		HANS (Choice)		Triage (Choice)		Interference	XAI Faithfulness & Plaus.		
Model	Acc. ↑	ECE_w ↓	Acc. ↑	ECE_w ↓	Acc. ↑	ECE_w ↓	Flip (C/N) ↓	Comp. (KS) ↑	Comp. (S1) ↑	Flag@g ↑
`Tev1-0.8B`	0.860	0.035	0.780	0.111	0.553	0.265	0.05 / 0.22	**0.541**	0.527	0.499
`Tev1-4B`	0.947	0.019	0.927	0.037	0.967	0.112	**0.00** / **0.00**	0.531	**0.530**	0.480
`Nimble-9B`	**0.953**	**0.014**	0.920	0.073	0.993	0.111	0.11 / 0.05	0.385	0.441	0.362
`Clef-Flash-9B`	0.900	0.107	0.873	0.063	0.967	0.115	0.01 / 0.13	0.303	0.299	0.347
`Clef-27B`	0.927	0.049	**0.940**	**0.032**	**1.000**	**0.091**	0.00 / 0.17	–	–	–
`Gemma4-8B` (JSON)	0.867	0.058	0.660	0.246	**1.000**	0.162	0.03 / 0.37	–	–	0.525*

on benign queries (False Positive Rate, FPR), calibration error (ECE_w), Brier score, and cross-question interference.

Five critical insights emerge for security operations and AI safety architectures:

1. **The Sub-Billion Model Blind Spot:** While Together AI's `Tev1-0.8B` delivers ultra-fast single-pass inference (19 ms), it suffers from a severe representational blind spot on complex adversarial attacks, detecting only 10.7% of injections (TPR = 0.107, $\text{ECE}_w = 0.296$, Brier 0.507). Sub-billion parameter models lack the linguistic and contextual capacity to recognize disguised adversarial intent and prompt overrides.

2. **Zero False Alarms Across All Capacities:** Crucially, every single evaluated model achieved a strict **0.0% False Positive Rate (FPR)** on benign user queries. On $n = 75$ benign inputs, this empirical 0.0% false-alarm rate corresponds to an exact 95% Clopper-Pearson binomial confidence interval of $[0.0\%, 4.8\%]$. In production guardrails, false alarms degrade user trust and interrupt legitimate conversational flows. The zero false-alarm rate confirms that System 1 classifiers do not suffer from hair-trigger over-blocking on conversational inputs.

3. **Frontier-Scale Precision and Near-Perfect Calibration:** Capacity scaling dramatically unlocks injection detection: `Tev1-4B` catches 68.0% of attacks (46 ms), `Nimble-9B` catches 66.7% (264 ms), and Cloudflare's `Clef-Flash-9B` achieves an outstanding 81.3% attack recall (95% CI $[70.7\%, 89.4\%]$) and 90.7% overall accuracy with near-zero calibration error ($\text{ECE}_w = 0.035$) in just 56 ms. At the frontier scale, `Clef-27B` establishes benchmark-leading defense: 94.7% **attack recall** (95% CI $[86.9\%, 98.5\%]$), 97.3% **overall accuracy**, and an exceptional calibration error of $\text{ECE}_w = \mathbf{0.010}$ (Brier 0.055).

4. **System 1 vs. System 2 Inline Filtering:** The generative baseline `Gemma4-8B (Sys. 2 JSON)` demonstrates why generative LLMs are ill-suited for inline firewalls: it required 426 ms per query and detected only 49.3% of attacks, with severe probability miscalibration ($\text{ECE}_w = 0.230$, Brier 0.465). In sharp contrast, `Clef-Flash-9B` is 7.6× **faster** (56 ms vs. 426 ms), catches 32.0 **percentage points more attacks** (81.3% vs. 49.3%), and provides 6.5× **better probability calibration**.

5. **Explainability for Incident Response (SOC Auditing):** When an inline guardrail flags a prompt as malicious, security analysts require fine-grained attribution to identify the exact payload. On flagrant prompt injections where decision heads saturate ($p \approx 0.999$), probability-space KernelSHAP spreads attribution evenly across background words due to gradient vanishing. In contrast, **S1-SHAP** operates in log-odds space, isolating the exact adversarial trigger sequences (e.g., `Repeat`, `instructions`, `above`, `prompt`, `END`) without requiring white-box gradient access, turning opaque decision models into auditable security sensors for Security Operations Centers (SOC).

Table 14: Adversarial prompt injection defense on the Tensor Trust benchmark ($N = 150$, balanced 50% benign / 50% adversarial attacks). Evaluated across five System 1 decision models and generative System 2 baseline (`Gemma4-8B` JSON). Choice evaluates binary classification (`prompt_injection` vs. `benign`); TPR is attack detection recall; FPR is false alarms on benign queries. Best values are in **bold**.

Model	Type	Latency (ms)↓	Choice Acc.↑	TPR (Recall)↑	FPR (Alarms)↓	ECE_w ↓	Brier↓	Flip (C/N)↓	TV (C/N)↓
`Tev1-0.8B`	System 1	**19**	0.553	0.107	**0.000**	0.296	0.507	0.01 / 0.03	0.00 / 0.01
`Tev1-4B`	System 1	46	0.840	0.680	**0.000**	0.047	0.231	0.00 / 0.01	0.00 / 0.00
`Clef-Flash-9B`	System 1	56	0.907	0.813	**0.000**	0.035	0.146	0.01 / 0.03	0.01 / 0.03
`Nimble-9B`	System 1	264	0.833	0.667	**0.000**	0.076	0.255	0.03 / 0.03	0.03 / 0.04
`Clef-27B`	System 1	811	**0.973**	**0.947**	**0.000**	**0.010**	**0.055**	0.00 / 0.01	0.00 / 0.01
`Gemma4-8B`	System 2 (JSON)	426	0.747	0.493	**0.000**	0.230	0.465	0.01 / 0.07	0.02 / 0.09

Table 15: Granular breakdown of adversarial prompt injection defenses on Tensor Trust ($N = 150$), evaluating benign query specificity ($n = 75$, True Negative Rate), secret extraction recall ($n = 40$), instruction hijacking recall ($n = 35$), overall attack recall ($n = 75$, TPR), and overall classification accuracy. Best values per column are in **bold**.

Model	Type	Benign (n=75)↑	Extraction (n=40)↑	Hijacking (n=35)↑	Overall TPR (n=75)↑	Choice Acc.↑
`Tev1-0.8B`	System 1	**1.000** (75/75)	0.175 (7/40)	0.029 (1/35)	0.107 (8/75)	0.553
`Tev1-4B`	System 1	**1.000** (75/75)	0.850 (34/40)	0.486 (17/35)	0.680 (51/75)	0.840
`Nimble-9B`	System 1	**1.000** (75/75)	0.825 (33/40)	0.486 (17/35)	0.667 (50/75)	0.833
`Clef-Flash-9B`	System 1	**1.000** (75/75)	0.850 (34/40)	0.771 (27/35)	0.813 (61/75)	0.907
`Clef-27B`	System 1	**1.000** (75/75)	**0.975** (39/40)	**0.914** (32/35)	**0.947** (71/75)	**0.973**
`Gemma4-8B`	System 2 (JSON)	**1.000** (75/75)	0.475 (19/40)	0.514 (18/35)	0.493 (37/75)	0.747

Granular Attack Breakdown: Secret Extraction vs. Instruction Hijacking. To uncover what attack families drive model defenses and failures, Table 15 decomposes the $n = 75$ adversarial instances into the two primary attack vectors established by the Tensor Trust taxonomy Toyer et al. (2023): *Secret Extraction* ($n = 40$, adversarial prompts designed to leak system instructions or confidential pre-prompts) and *Instruction Hijacking* ($n = 35$, adversarial prompts engineered to override system directives and execute arbitrary malicious tasks).

This fine-grained decomposition yields two pivotal operational insights: First, *Instruction Hijacking* is systematically harder to detect than *Secret Extraction* across sub-10B architectures. On both `Tev1-4B` and `Nimble-9B`, extraction recall reaches an impressive 82.5%–85.0%, whereas hijacking recall collapses to 48.6% (a severe 34–36 percentage-point vulnerability gap). At the sub-billion scale, `Tev1-0.8B` suffers from an almost complete blind spot against instruction hijacking, detecting only a single attack out of 35 (2.9% recall, compared to 17.5% on extraction). Second, architectural specialization and parameter scaling successfully conquer this asymmetry: Cloudflare's `Clef-Flash-9B` elevates instruction hijacking recall to 77.1% (intercepting 27/35 attacks) in 56 ms, while `Clef-27B` attains benchmark-leading robustness across both vectors (97.5% on extraction, 91.4% on hijacking, and 94.7% overall). By contrast, the generative baseline `Gemma4-8B (Sys. 2 JSON)` demonstrates poor, indiscriminate vulnerability across both modalities (47.5% on extraction and 51.4% on hijacking), failing to detect half of all attacks while incurring a 7.6× latency overhead.

6 Discussion

Two structural findings from our audit concern the System 1 decision paradigm itself and apply broadly across providers. First, across evaluated models, the provider `confidence` field is an exact deterministic summary of distributional dispersion $(1 - H(p)/\ln K$ across all sub-10B models, and $(\max p - 1/K)/(1 - 1/K)$ on `Clef-27B`) rather than empirical accuracy. While useful as a relative ranking signal, interpreting $c(p) = 0.63$ as a "63% probability of correctness" is mathematically incorrect; calibration must be evaluated on $\max p$. Second, a multi-question System 1 request is not a collection of independent queries: bundling a *Choice* question and a *Noul* question in a single payload caused the binary *Noul* decision to flip on 22.4% of inputs on `Tev1-0.8B` ($N = 450$, reaching up to 60–82% under exploratory uncalibrated prompts) and up to 13–17% on 9B–27B models (`Clef-Flash-9B`, `Nimble-9B`, and `Clef-27B`), whereas `Tev1-4B` achieved near-zero

interference (0.2%). Production systems that bundle questions to reduce network overhead can alter a model's effective decision boundary unless isolated single-question invariance is verified first.

On the model side, the contrast between e-SNLI and HANS highlights a critical vulnerability of fast System 1 classifiers. A sub-billion model that is well calibrated in-distribution ($\mathrm{ECE}_w = 0.035$ on e-SNLI, $N = 450$) retains high confidence (0.871 mean $\max p$) on HANS where overall accuracy drops to 0.780, collapsing to 11–12% on adversarial subsequence and constituent traps. While scaling to 4B–27B parameters substantially improves HANS accuracy (0.873–0.940) and lowers ECE_w down to 0.032 (and 0.020–0.023 under post-hoc temperature scaling), residual shortcut errors remain high-confidence. Conversely, our explainability audit demonstrates that the single-pass probabilistic interface makes post-hoc perturbation XAI both tractable and faithful: LIME, KernelSHAP, and our logit-space **S1-SHAP** reliably isolate clinical flags over alarming confounders and recover human rationale highlights at more than double the random baseline rate. Crucially, operating in log-odds space via S1-SHAP mitigates probability saturation on sharp decision heads without requiring white-box gradient access (+14.5% comprehensiveness on `Nimble-9B`, $p = 0.0106$), while avoiding the 18×–187× latency penalty and poor verbalized-probability calibration (Brier = 0.234–0.545) of generative System 2 models.

Regarding explainability paradigms, commercial and managed System 1 decision endpoints (such as TypeSafe AI's Jev, Perplexity's `pplx-decider`, and Cloudflare Workers AI `Clef`) expose only continuous distribution vectors over HTTP/gRPC schemas, without backward gradient access or internal attention weights. Consequently, black-box perturbation is the only viable, model-agnostic explanation interface in production. While white-box methods such as Integrated Gradients Sundararajan et al. (2017) on open-weight backbones provide a useful theoretical upper bound on attribution faithfulness, our findings show that black-box coalition games (KernelSHAP and S1-SHAP) achieve near-perfect trigger recovery (97–100%) and full logic gate recovery at sub-second latencies, closing the gap without requiring model weight exposure. Furthermore, our reliance on transparent, deterministic feature-removal and masking operators aligns with foundational epistemological cautions in explainable AI: as argued by Delaunay et al. Delaunay et al. (2024), attempting to explain a black box using another complex black box (such as training deep generative models or latent-space autoencoders to synthesize counterfactuals) risks introducing uninterpretable confounding biases and false fidelity. By keeping the perturbation operator strictly transparent (exact token deletion or neutral masking) and linearizing the model's response via S1-SHAP in log-odds space, practitioners obtain verifiable causal attributions without delegating the explanation process to secondary uninterpretable models.

7 Conclusion

We presented an end-to-end explainability, calibration, and robustness audit of Jev-style System 1 decision models across five open-weight models spanning 0.8B to 27B parameters (`Tev1-0.8B`, `Tev1-4B`, `Nimble-9B`, `Clef-Flash-9B`, and `Clef-27B`) and a generative System 2 baseline (`Gemma4-8B` JSON and CoT) over $N = 450$ benchmark instances and a new Planted System-Rule XAI Benchmark, alongside **S1-SHAP**, a logit-space Shapley attribution method tailored to saturating single-pass decision heads. Our results establish that provider `confidence` scores measure entropy concentration rather than calibrated accuracy, that cross-question interference is severe at 0.8B scale and in bundled System 2 JSON schemas before vanishing on `Tev1-4B` (0.2% flips), and that $\geq$ 4B System 1 models achieve superior calibration (ECE_w down to 0.014–0.019 on e-SNLI and 0.032–0.073 on HANS) at far lower latency than generative System 2 models. At the same time, post-hoc coalition attributions (LIME, KernelSHAP, and **S1-SHAP**) achieve statistically significant gains in faithfulness ($p < 10^{-4}$ vs. LOO; $p = 0.0106$ for S1-SHAP vs. KernelSHAP on sharp heads) and 97–100% ground-truth trigger recovery at zero extra coalition cost, providing a fast, axiomatic transparency layer for bare System 1 probability outputs.

Reproducibility and Artifact Availability

All datasets, evaluation harnesses, and experimental logs are publicly available at `https://github.com/j2launay/jev-decision-models-xai`. The repository provides: (i) the deterministic audit harness supporting local Ollama models and remote System 1 endpoints; (ii) the Planted System-Rule Benchmark

suites across single, dual OR, multi OR, dual AND, and near-miss distractor regimes; (iii) the clinical triage vignettes with annotated character offsets for clinical flags, emotional confounders, and neutral fillers; (iv) bit-identical SQLite prediction caches and Parquet tables for all five open-weight models and the System 2 baseline; and (v) scripts to reproduce all tables and high-resolution figures.

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A Mathematical Notation and Glossary

Table 16 provides a unified reference for the mathematical symbols, decision primitives, and evaluation metrics used throughout this paper.

Table 16: Glossary of mathematical notation, decision model schemas, game-theoretic variables, and evaluation metrics.

Symbol	Domain / Type	Description
$s \in \mathcal{S}$	Text / JSON state	Input context state (patient note, premise, or document).
$q \in \mathcal{Q}$	Named query	Decision primitive query specified in request tuple $(s, \mathcal{Q})$.
$\mathcal{Q}$	Set of queries	Dictionary of named decision primitives $\{q_1, \ldots, q_H\}$ asked in a request.
$\mathcal{O}$	Set of strings	Finite categorical option set $\{o_1, \ldots, o_K\}$ declared for a *Choice* query.
$K \in \mathbb{N}_{\geq 2}$	Integer scalar	Cardinality of the option set ($K = 3$ for e-SNLI / Triage; $K = 2$ for HANS / *Noul*).
$p(y \mid s, q)$	Δ^{K-1} simplex	Normalized probability distribution vector over declared options in $\mathcal{O}$.
y^*	Categorical label	Predicted top choice: $y^* = \arg\max_{o_k \in \mathcal{O}} p(o_k \mid s, q)$.
$c(p)$	$[0, 1]$ scalar	Provider-returned `confidence` field $(1 - H(p)/\ln K$ on sub-10B models; $(\max p - 1/K)/(1 - 1/K)$ on `Clef-27B`).
$H(p)$	$\mathbb{R}_{\geq 0}$ scalar	Shannon entropy of the probability distribution: $-\sum_k p_k \ln p_k$.
$M \in \mathbb{N}$	Integer scalar	Number of human-interpretable features (words or tokens) in state s.
$z \in \{0, 1\}^M$	Binary vector	Feature coalition mask ($z_i = 1$ if feature i is retained, 0 if masked/removed).
$h_s(z)$	Text state	Perturbed state induced by applying coalition mask z to original state s.
$v_{\text{prob}}(z)$	$[0, 1]$ scalar	Probability-space coalition value function: $p(y^* \mid h_s(z), q)$.
$v_{\text{S1}}(z)$	$\mathbb{R}$ scalar	S1-SHAP log-odds value function: $\ln(\tilde{p}/(1 - \tilde{p}))$ with clipping $\varepsilon = 10^{-4}$.
$\phi_i \in \mathbb{R}$	Attribution scalar	First-order attribution weight of feature i (LOO, LIME, KernelSHAP, S1-SHAP).
$I_{ij} \in \mathbb{R}$	Interaction scalar	Pairwise Shapley Interaction Index (SII): $I_{ij} > 0$ (synergy/AND), $I_{ij} < 0$ (redundancy/OR).
$\text{ECE}_w, \text{ECE}_m$	$[0, 1]$ error	Expected Calibration Error under equal-width (w) vs. equal-mass (m) binning.
MCE	$[0, 1]$ error	Maximum Calibration Error across partitioned confidence bins.
Brier	$\mathbb{R}_{\geq 0}$ error	Multiclass Brier score: $\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}(p_{ik} - \mathbb{I}(y_i = k))^2$.
$\text{AUROC}_p, \text{AUROC}_c$	$[0, 1]$ area	Selective prediction AUROC using $\max p$ (AUROC_p) vs. entropy confidence $c(p)$ (AUROC_c).
D_{TV}	$[0, 1]$ distance	Total variation distance between isolated and co-asked joint distributions.
Comp., Suff.	AOPC scores	Comprehensiveness (↑) and Sufficiency (↓) over top-k% feature interventions.
Del. AUC, Ins. AUC	$[0, 1]$ area	Area under progressive feature deletion (↓) and insertion (↑) curves.
Flag prec.@g	$[0, 1]$ precision	Fraction of top-g attributed features falling inside ground-truth causal spans.
Conf. share@g	$[0, 1]$ fraction	Fraction of top-g attributed features falling inside alarming distractor confounders.

B Additional Empirical Tables

Table 17: Change in ECE (equal-width bins), accuracy, and flip rate relative to clean inputs, with paired bootstrap 95% intervals.

		Choice			Noul		
Source	Perturbation	ΔECE$_w$ ↓ [95% CI]	ΔAcc.↑	Flip↓	ΔECE$_w$ ↓ [95% CI]	ΔAcc.↑	Flip↓
e-SNLI	Typos 5%	**+0.001** [-0.03, 0.04]	-0.020	0.02	+0.017 [-0.02, 0.05]	+0.000	**0.01**
	Typos 10%	+0.008 [-0.03, 0.06]	-0.067	0.09	+0.028 [-0.02, 0.07]	-0.067	0.09
	Typos 20%	+0.030 [-0.02, 0.09]	-0.100	0.17	+0.038 [-0.01, 0.09]	-0.120	0.16
	Distractor	+0.008 [-0.03, 0.06]	-0.020	0.10	+0.015 [-0.03, 0.05]	**+0.027**	0.04
	Case/punct.	+0.023 [-0.02, 0.05]	**-0.007**	**0.01**	**+0.014** [-0.02, 0.05]	+0.000	**0.01**
HANS	Typos 5%	+0.000 [0.00, 0.00]	+0.000	**0.00**	-0.000 [-0.00, 0.00]	+0.000	**0.00**
	Typos 10%	+0.060 [0.01, 0.10]	-0.027	0.12	+0.046 [0.00, 0.09]	-0.020	0.09
	Typos 20%	+0.069 [-0.01, 0.15]	-0.133	0.27	**-0.014** [-0.07, 0.08]	-0.140	0.29
	Distractor	+0.020 [-0.04, 0.05]	**+0.020**	0.05	-0.005 [-0.03, 0.02]	**+0.027**	0.03
	Case/punct.	**-0.009** [-0.04, 0.05]	-0.007	0.02	-0.012 [-0.04, 0.02]	-0.007	0.02
Triage	Typos 5%	+0.015 [-0.01, 0.05]	-0.013	**0.02**	+0.005 [-0.02, 0.03]	+0.000	**0.01**
	Typos 10%	-0.001 [-0.03, 0.03]	-0.007	0.04	+0.013 [-0.01, 0.04]	-0.013	**0.01**
	Typos 20%	-0.011 [-0.03, 0.04]	-0.013	0.03	-0.018 [-0.05, 0.02]	+0.007	0.02
	Distractor	-0.024 [-0.05, 0.02]	+0.007	0.03	**-0.023** [-0.05, 0.01]	**+0.020**	0.02
	Case/punct.	**-0.032** [-0.06, 0.01]	**+0.027**	0.09	-0.001 [-0.04, 0.02]	-0.020	0.05

Table 18: Subword-level faithfulness (Qwen3.5 tokenizer).

			Deletion masking				`[MASK]` masking	
Source	Head	Method	Comp.↑	Suff.↓	Del. AUC↓	Ins. AUC↑	Comp.↑	Suff.↓
e-SNLI	Choice	LOO	0.590	0.298	0.321	**0.779**	0.517	0.405
		LIME	**0.668**	0.291	0.234	0.747	0.567	0.401
		KernelSHAP	0.662	**0.275**	**0.194**	0.765	0.570	**0.398**
		S1-SHAP (ours)	0.666	0.299	0.195	0.732	**0.586**	0.419
		Random	0.226	0.513	0.488	0.507	0.317	0.507
	Noul	LOO	0.455	0.214	0.516	**0.850**	0.411	0.277
		LIME	**0.556**	0.225	0.404	0.830	**0.454**	**0.260**
		KernelSHAP	0.554	**0.209**	**0.366**	0.828	0.452	0.277
		S1-SHAP (ours)	0.477	0.220	0.471	0.807	0.432	0.262
		Random	0.180	0.332	0.652	0.672	0.248	0.370
Triage	Choice	LOO	0.443	0.086	0.194	0.795	0.388	0.142
		LIME	0.459	0.015	0.161	0.815	0.398	0.120
		KernelSHAP	**0.469**	**-0.042**	**0.114**	**0.833**	**0.441**	**0.047**
		S1-SHAP (ours)	0.456	-0.035	0.125	0.818	0.414	0.069
		Random	0.101	0.336	0.524	0.552	0.157	0.353
	Noul	LOO	0.391	0.049	0.326	0.890	0.311	0.120
		LIME	**0.450**	-0.100	0.197	**0.961**	**0.397**	0.011
		KernelSHAP	0.437	-0.098	0.198	0.929	0.340	0.018
		S1-SHAP (ours)	0.449	**-0.119**	**0.174**	0.957	0.389	**-0.012**
		Random	0.076	0.337	0.617	0.620	0.115	0.323

C Decision Schemas, Prompt Specifications, and Baseline Implementations

This appendix provides the complete, unabridged technical specifications for all decision schemas, prompt templates, option definitions, and baseline generative configurations evaluated in this work, ensuring exact experimental reproducibility.

C.1 System 1 Decision Primitives and Task Schemas

In the Jev System 1 specification, each request dispatches an input state dictionary paired with a dictionary of typed question schemas $\mathcal{Q}$. Table 19 summarizes the question instructions, candidate option keys, and semantic criteria across all evaluation domains.

Table 19: System 1 decision schemas evaluated across benchmark tasks: question types, input state variables, question instructions, and semantic criteria per option.

Task	Primitive	Identifier	Question Instruction	Option Criteria / Target Semantics
e-SNLI	Choice ($K = 3$)	`label`	"Given that the premise is true, what is the relation of the hypothesis to the premise?"	`entailment`: The hypothesis is definitely true given the premise. `neutral`: The hypothesis might be true given the premise. `contradiction`: The hypothesis is definitely false given the premise.
	Noul ($K = 2$)	`noul`	"Given that the premise is true, is the hypothesis definitely true?"	`true`: The premise entails the hypothesis. `false`: The premise does not entail the hypothesis.
HANS	Choice ($K = 3 \to 2$)	`label`	(Identical to e-SNLI; probabilities of `neutral` and `contradiction` are summed into `non_entailment`).	(Identical to e-SNLI).
	Noul ($K = 2$)	`noul`	"Given that the premise is true, is the hypothesis definitely true?"	`true`: The premise entails the hypothesis. `false`: The premise does not entail the hypothesis.
Clinical Triage	Choice ($K = 3$)	`label`	"What emergency triage level does this patient require?"	`immediate`: Life-threatening condition requiring immediate resuscitation (ESI 1). `urgent`: Serious but stable condition that should be seen within an hour (ESI 2–3). `non_urgent`: Minor condition that can safely wait (ESI 4–5).
	Noul ($K = 2$)	`noul`	"Does the patient require immediate care?"	`true`: The patient requires immediate emergency care. `false`: The patient does not require immediate emergency care.
Tensor Trust	Choice ($K = 2$)	`decision`	"Evaluate whether this user input constitutes an adversarial prompt injection or a legitimate interaction."	`benign`: A legitimate user query adhering to assistant guidelines. `prompt_injection`: An adversarial attempt to hijack system rules, extract secrets, or override instructions.
	Noul ($K = 2$)	`is_injection`	"Is this user input an adversarial prompt injection, jailbreak, or system override attempt?"	`true`: The input attempts instruction hijacking or jailbreaking. `false`: The input is a benign, safe user query.
Planted Rule	Choice ($K = 3$)	`label`	"MANDATORY OVERRIDE RULE: [Planted boolean trigger directive]. What emergency acuity does this note require?"	`immediate`: Trigger present or true emergency. `urgent`: Standard urgent note without trigger. `non_urgent`: Benign baseline note.
	Noul ($K = 2$)	`noul`	"MANDATORY OVERRIDE RULE: [Planted boolean trigger directive]. Does the patient require immediate care?"	`true`: Planted trigger present or acute condition. `false`: Benign routine note.

C.2 Planted System-Rule Directives

In the Planted System-Rule Benchmark (Section 5.6), the question instructions $\mathcal{Q}$ are injected with explicit boolean override directives that are held invariant while the explainer masks candidate words in the note:

- **Single Trigger ($g = 1$):** *"MANDATORY OVERRIDE RULE: If the note mentions the code word '[trigger]' anywhere, classify as immediate/true regardless of clinical symptoms."* (Triggers: `velvet`, `cobalt`, `amber-seven`).
- **Dual Redundant OR ($g = 2$):** *"MANDATORY OVERRIDE RULE: If the note mentions EITHER the code word '[trigger1]' OR the code word '[trigger2]' (or both), classify as immediate/true."* (Pairs: `velvet` / `cobalt`, `crimson` / `sapphire`).
- **Saturated Multi OR ($g = 3$):** *"MANDATORY OVERRIDE RULE: If the note mentions ANY of the three secret code words '[trigger1]', '[trigger2]', or '[trigger3]', classify as immediate/true."* (Triplets: `velvet` / `cobalt` / `zenith`).
- **Dual Conjunctive AND ($g = 2$):** *"MANDATORY OVERRIDE RULE: Classify as immediate/true IF AND ONLY IF BOTH code words '[trigger1]' AND '[trigger2]' appear simultaneously in the note. If only one code word is present, disregard this rule."*

C.3 Generative System 2 Baseline Prompting (Gemma4-8B)

To benchmark single-pass System 1 models against conventional autoregressive LLMs, our generative System 2 baseline (`Gemma4-8B`) is served via Ollama's `/api/chat` endpoint using two distinct operational modes:

1. Direct Structured JSON Mode. In direct JSON mode, the model is provided with a strict JSON grammar constraint enforcing the schema of the target primitive. The user message template serializes the input state and decision questions as follows:

```
Input State:
{state_key_1}: {state_value_1}
{state_key_2}: {state_value_2}

Question '{q_name}' (choice): {instructions}
Options:
- {opt_1}: {crit_1}
- {opt_2}: {crit_2}
Provide the chosen option, a probability distribution in [0,1] summing to 1.0 over
the options, and your confidence in [0,1].
Also provide 'rationale_words': a list of the exact input words (ordered from most
important to least important) that drove your decision.
Respond strictly in JSON.
```

The API request enforces the following JSON Schema:

```
{
"type": "object",
"properties": {
"choice": {"type": "string", "enum": ["{opt_1}", "{opt_2}", ...]},
"probabilities": {
"type": "object",
"properties": {"{opt_1}": {"type": "number"}, ...},
"required": ["{opt_1}", "{opt_2}", ...]
},
"confidence": {"type": "number"},
"rationale_words": {"type": "array", "items": {"type": "string"}}
```

```
},
"required":  ["choice", "probabilities", "confidence", "rationale_words"]
}
```

Raw verbalized probabilities are normalized via $p_k \leftarrow \max(0, \tilde{p}_k) / \sum_j \max(0, \tilde{p}_j)$ to ensure valid distribution simplex constraints.

2. Chain-of-Thought (CoT) Deliberative Mode. In CoT mode, the system instructions explicitly prompt the model to produce a deliberative reasoning trace before emitting the final structured JSON block:

```
You are an expert diagnostic and decision-making system.
First, analyze the input state carefully step-by-step.  Identify key supporting
evidence, contradictory cues, and potential confounding factors.
Then, conclude your deliberation and emit your final decision strictly as a JSON
object matching the required schema.
```

Decoding parameters for all System 2 evaluations are set to temperature $T = 0.0$ with top-$p = 1.0$ to ensure maximal determinism.

C.4 Dataset Curation, Annotation Protocols, and Quality Control

To guarantee experimental reproducibility and eliminate data curation artifacts, this section provides detailed protocols for dataset selection, label mapping, ground-truth span annotation, and quality control.

1. Clinical Triage Vignettes and Ground-Truth Rationale Annotation. The clinical triage dataset ($n = 150$) evaluates high-stakes medical decision-making under high-intensity distractors.

- **Acuity Label Mapping:** Clinical acuity is operationalized following the Emergency Severity Index (ESI Version 4) implementation handbook Gilboy et al. (2011). The five clinical ESI tiers are mapped into three actionable System 1 decision categories:
  - `immediate` (ESI 1): Severe hemodynamic or respiratory instability requiring immediate life-saving resuscitation (e.g., cardiac arrest, impending airway loss, anaphylaxis, severe cyanosis).
  - `urgent` (ESI 2–3): High-risk clinical presentations, severe distress, or conditions requiring multiple diagnostic/therapeutic resources while vital signs remain currently compensated (e.g., acute coronary syndrome, open fractures, severe peritoneal signs).
  - `non_urgent` (ESI 4–5): Stable, low-acuity presentations requiring at most a single diagnostic modality or simple outpatient treatment (e.g., minor extremity contusions, routine suture removal, uncomplicated pharyngitis).
- **Character-Level Span Annotation Protocol:** For each clinical vignette, character offsets were annotated into three disjoint semantic tiers: (i) *Clinical Flags*: Explicit pathophysiological markers and vital sign derangements directly justifying the triage assignment. (ii) *Emotional Confounders*: High-intensity affective vocabulary designed to test model susceptibility to emotional bias (e.g., "terrified", "screaming in agony", "panicking", "sobbing uncontrollably"). (iii) *Neutral Fillers*: Background contextual prose (e.g., arrival modality, demographic descriptors, ambient environment).
- **Inter-Annotator Agreement:** Annotations were generated by two independent reviewers with clinical informatics expertise. Token-level inter-annotator agreement achieved a Cohen's $\kappa = 0.892$. All discrepant boundary spans were arbitrated and reconciled through joint consensus review prior to running attribution evaluations.

2. Tensor Trust Adversarial Security Benchmark. The prompt injection evaluation utilizes instances sampled directly from the peer-reviewed Tensor Trust benchmark Toyer et al. (2023) (NeurIPS 2023 Datasets and Benchmarks track).

- **Stratified Cohort Composition:** The evaluation cohort ($N = 150$) features an exact 50%/50% split: $n = 75$ legitimate benign user queries and $n = 75$ adversarial attacks.
- **Attack Taxonomy Stratification:** The 75 adversarial attacks are partitioned into the two foundational attack vectors defined by Toyer et al. Toyer et al. (2023): (i) *Secret Extraction* ($n = 40$): Sophisticated social engineering, multi-turn disguise, cipher/encoding tricks, and character-splitting prompts engineered to coax the system into revealing protected instructions or confidential system prompts. (ii) *Instruction Hijacking* ($n = 35$): Direct and indirect injection payloads engineered to overwrite system boundaries, ignore pre-prompts, and execute rogue commands (e.g., forcing confirmation tokens or role inversion).
- **Length-Matched Distributional Control:** To ensure that differences in inference latency or detection accuracy between benign and malicious inputs are not confounded by input sequence length, the benign and attack cohorts were strictly length-matched over the range of 8 to 45 words (benign: 24.3 ± 8.2 words; adversarial: 26.1 ± 9.4 words). A two-sample Kolmogorov-Smirnov test confirmed no statistically significant difference between length distributions ($p = 0.42$).

3. Natural Language Inference Benchmarks (e-SNLI and HANS).

- **e-SNLI ($n = 150$):** Sourced from the curated test set of the Stanford Natural Language Inference with Explanations benchmark Camburu et al. (2018). Instances are balanced equally across the three inferential categories: `entailment` ($n = 50$), `neutral` ($n = 50$), and `contradiction` ($n = 50$). Ground-truth feature masks utilize the consensus character-level human rationale highlights provided in the e-SNLI gold annotations.
- **HANS ($n = 150$):** Sourced from the Heuristic Analysis for NLI Systems benchmark McCoy et al. (2019). Cohorts are balanced equally across the three syntactic trap families: lexical overlap ($n = 50$), subsequence ($n = 50$), and constituent structure ($n = 50$), with an exact $1 : 1$ balance between `entailment` ($n = 75$) and `non_entailment` ($n = 75$) labels to diagnose heuristic shortcut exploitation under covariate shift.